Short-selling constraints and interacting asset prices

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Abstract

We study two interacting asset markets when heterogeneous beliefs are determined by evolutionary competition and short selling is prohibited. We provide conditions on beliefs such that short-selling constraints are active or slack in equilibrium for an arbitrary number of investor types. We show how to use these active-slack analytics to obtain fast solutions to models with large-scale heterogeneity. Two dynamic asset pricing applications show the utility of these results: we obtain fast solutions and distinct price dynamics when many predictors are in evolutionary competition. We find selective shorting bans reduce mispricing relative to blanket bans on both assets.

1 Introduction

In a series of papers, Lintner (1969, 1971, 1975) shows that when investors are heterogeneous and short-selling is possible, the traditional form of the capital asset pricing model remains intact (under certain conditions) and there is no price effect of short sales; further, this result holds even with margin requirements for short selling. However, as Miller (1977) and Ross (1977) point out, this equivalence result – and the corresponding invariance of prices – does not hold when investors are heterogeneous and face short-selling prohibitions. Unfortunately, pricing risky assets with heterogeneous investors and short-selling constraints is difficult.

In this paper we take on this problem of equilibrium asset pricing with heterogeneous, short-selling constrained investors. It is well-known that such problems are, in general, both analytically and computationally intractable. On the one hand, checking the Karush-Kuhn-Tucker conditions requires enumerating all possible scenarios when short-selling constraints become active, which is analytically intractable aside from simple cases where the size of the problem is small. On the other hand, even advanced computational methods may be inadequate for obtaining solutions of large problems due to the curse of dimensionality that arises from the combination of many investors and many short-selling constraints.

We consider a problem with two risky assets and an arbitrary number of investors. Our main contribution is to tame the curse of dimensionality in this setting by presenting cases of economic interest where finding an equilibrium is analytically and computationally tractable

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even with large-scale heterogeneity. Our main insight is that, under certain conditions, ranking investors types in terms of *belief optimism* allows many active-constraint scenarios to be ruled out *a priori*, so that the computational burden of large-scale heterogeneity is drastically reduced. We illustrate our approach using analytical and numerical examples. Since our results are built on analytics, our solution algorithms are fast and avoid approximation.

Our starting point is the Jarrow (1980) capital asset pricing model in which investors are heterogeneous and are prohibited from short selling. We study a two-asset, many-investor version of the model for analytic tractability. In contrast to Jarrow's static model, we extend the model with *dynamic* asset pricing, and belief adoption responds to past performance via evolutionary competition (Brock and Hommes, 1997, 1998). Thus, relative to past work, we extend asset pricing with evolutionary competition and short-selling constraints to *two* risky assets whose prices are jointly determined – a difficult fixed point problem.[1] The curse of dimensionality is exacerbated in a dynamic model as studied here, but our approach – based on ordering investors in terms of optimism – cuts the curse of dimensionality down to size.

The first case we study is that of a short-selling ban on *one asset*, which is analogous to selective bans on shorting certain shares or sectors (financial stocks) as in the Financial Crisis of 2008-9 (see Siciliano and Ventoruzzo, 2020), rather than a blanket market-wide ban. In this case, ordering investors in terms of optimism facilitates a fast solution because the number of cases is cut from exponential with base 2 to *linear* in the number of types. Our second case combines a shorting ban on both risky assets with *aligned optimism*, where optimism refers to beliefs about both assets. We consider both positively and negatively aligned beliefs. With positive alignment an investor's position in the optimism ranking is the same for both risky assets, e.g. investors who use a common extrapolation coefficient (heuristic) to forecast future prices of over- or under-valued assets. With negative alignment, an investor's optimism ranking is *reversed* for each asset, reflecting sharp investor disagreement about asset values (e.g. fundamental versus fiat assets or price extrapolation for under- versus over-priced assets). In this case, ordering investors reduces the number of cases from exponential with base 4 to *quadratic* in the number of investor types – a dramatic improvement.

To illustrate the utility of these results, we present analytic examples as well as numerical applications that test the speed of solution algorithms built on our analytical results and study the economic implications of short-selling bans in many-investor settings. Our algorithms are fast and accurate as expected. Our numerical examples show that large-scale heterogeneity can lead to materially different results than an otherwise identical small-investors world, including different conclusions on the merits of short-selling restrictions. In particular, we find that selective short-selling bans may reduce mispricing relative to blanket shorting bans on both assets, suggesting the need for careful policy analysis and evaluation.

[1] One risky asset and arbitrarily many types in evolutionary competition is studied in Brock et al. (2005). Short-selling constraints are added in Anufriev and Tuinstra (2013) and Hatcher (2024a,b); note that even the one-asset case presents considerable difficulties because the market-clearing price depends on aggregate demand, but individual demands are 'kinked' (i.e. piecewise-linear) at investor-specific price thresholds.

2 Preliminaries

Following Jarrow (1980), we consider a capital asset pricing model where short selling is banned. We extend the model with dynamics and allow belief updating (Section 3.2).

Our benchmark assumptions are as follows:

A.1 There are no transactions costs or taxes, and asset shares are infinitely divisible. Buyers and sellers act as price takers, and all risky assets are in positive outside supply.

A.2 Riskless rate is exogenous. There is unrestricted borrowing and lending at this rate.

A.3 Investors are myopic and they maximize the expected utility of future wealth (one period ahead). Risk aversion and non-satiation is assumed.

A.4 Investors have mean-variance utility with heterogeneous risk aversion coefficients.

A.5 Investors have heterogeneous expectations of the mean vector and covariance matrix of next period's risky asset prices. Each investor's covariance matrix is positive definite.

A.6 Short selling of risky assets is prohibited.

Assumptions (A.1)–(A.3) are standard in the literature. By assumption (A.2), investors have access to liquidity and they do not face any leverage constraints. The assumption of mean-variance utility in (A.4) is fairly standard and it is well-known that maximizing such an objective function is equivalent to maximizing expected utility when investors have constant absolute risk aversion utility functions and asset returns follow a normal distribution.

Investors enter the first period with given initial wealth and their expectations; they take the market price as given. Given their expectations, investors choose demands that maximize their utility, subject to short-selling constraints. The demands, in turn, determine risky asset prices and investor positions, which is the market equilibrium of interest. After the current asset price has been realized, investors form expectations (i.e. forecasts) about asset prices in the next period and a new equilibrium price vector and asset allocation is determined. This process repeats with expectations, portfolios choices and price updated each period. Note that investors with identical expectations and risk aversion but different initial wealth would have the same asset holdings given unrestricted positions in the riskless asset (A.3); therefore, having heterogeneity that extends beyond wealth (A.4, A.5) is crucial.[2]

The model can easily be extended to include some aspects (dividends, heterogeneous weights on current prices) that are absent in the main model. We set out these extensions in the Supplementary Appendix (here); see also Section 5. In addition to assumptions A.1 to A.6 – which also appear in Jarrow (1980) – we also make some assumptions about investors' dynamic belief updating and the evolution of population shares that use each forecasting rule. We discuss these extra assumptions once the portfolio problem has been introduced.

[2]Intuitively, differences in wealth are not a constraint provided investors can take unrestricted positions in the riskless asset. Therefore, differences in positions arise due to differences in expectations or risk aversion.

3 Model

We follow closely the notation in Jarrow (1980). There are two risky assets with date t prices $P_{1,t}, P_{2,t}$. There is a finite number of periods, $T \in \mathbb{N}_{>0}$. The riskless asset is security 0; its return is fixed at $1 + r > 1$ and its price is exogenous; we set $P_{0,t} = 1$ for convenience.

3.1 Investor problem

There are K myopic investor types. At each date $t \in [T]$, each investor type $k \in [K]$ solves a constrained portfolio selection problem given their individual beliefs over future wealth:

$$\max_{N^k_{0,t}, N^k_{1,t}, N^k_{2,t}} \quad \mathbb{E}^k_t W^k_{t+1} - \frac{a^k}{2} \mathrm{Var}^k_t[W^k_{t+1}] \tag{1}$$

subject to

$$W^k_{t+1} = \sum_{j=1}^{2} N^k_{j,t} P_{j,t+1} + (1+r)\left(W^k_t - \sum_{j=1}^{2} N^k_{j,t} P_{j,t}\right)$$

$$N^k_{j,t} \geq 0, \quad j = 1, 2$$

where $a^k > 0$ is the risk aversion coefficient of investor type k, and $N^k_{0,t} = W^k_t - \sum_{j=1}^{2} N^k_{j,t} P_{j,t}$ is their (unrestricted) position in the riskless asset, 0.

Equation (1) shows that each investor maximizes mean-variance utility over future wealth. The superscript k allows each investor to have (potentially) distinct expectations as reflected in the expectations operator $\mathbb{E}^k_t$ and the variance operator Var^k_t. The number of shares of asset j demanded by investor k is $N^k_{j,t}$; given short-selling restrictions, $N^k_{1,t}, N^k_{2,t} \geq 0 \ \forall t, k$. The fixed outside supplies of the risky assets are strictly positive: $\overline{N}_1, \overline{N}_2 > 0$.

Given (1), investor type k's problem is equivalent to

$$\max_{N^k_{1,t}, N^k_{2,t} \geq 0} \ \sum_{j=1}^{2} N^k_{j,t}(\mathbb{E}^k_t P_{j,t+1} - (1+r)P_{j,t}) + (1+r)W^k_t - \frac{a^k}{2} \sum_{i=1}^{2} \sum_{j=1}^{2} N^k_i N^k_j \sigma^k_{ij} \tag{2}$$

where $\mathbb{E}^k P_{j,t+1}$ is k's expectation (or forecast) of $P_{j,t+1}$, and $\sigma^k_{i,j}(t)$ reflects their date t belief of the price covariance of assets i and j; note the constraints $N^k_{j,t} \geq 0$ are reflected in (2).

The Lagrangian of (2) yields the following first-order conditions for each type $k \in [K]$:

$$\lambda^k_{j,t} N^k_{j,t} = 0, \qquad \lambda^k_{j,t} \geq 0, \qquad N^k_{j,t} \geq, \qquad j = 1, 2 \tag{3}$$

$$\mathbb{E}^k_t P_{j,t+1} - (1+r)P_{j,t} + \lambda^k_{j,t} = a^k \sum_{i=1}^{2} N^k_{i,t} \sigma^k_{ij}(t), \quad j = 1, 2 \tag{4}$$

where $\lambda^k_{1,t}, \lambda^k_{2,t}$ are the Lagrange multipliers on the short-selling restrictions at date t.

The equations in (3) are the Kuhn-Tucker conditions. If the Lagrange multiplier satisfies

$\lambda^k_{j,t} = 0$ the short-selling constraint on asset j is slack at date t and $N^k_{j,t} \geq 0$. If $\lambda^k_{j,t} > 0$, the constraint is binding and $N^k_{j,t} = 0$. The system of equations in (4) defines k's optimal portfolio of risky assets, given their expectations and the short-selling constraints in (3).

The equations in (4) can be solved for the following demand schedules:

$$N^k_{j,t} = \sum_{i=1}^{2} v^k_{ji}(t)(\mathbb{E}^k_t P_{i,t+1} - (1+r)P_{i,t}) + \sum_{i=1}^{2} v^k_{ji}(t)\lambda^k_{i,t}, \qquad j = 1,2 \tag{5}$$

where v^k_{ij} is the ijth entry of $\mathbf{V}^k(t) := (a^k)^{-1}(\mathbf{\Omega}^k(t))^{-1}$ and $\mathbf{\Omega}^k(t) = [\sigma_{ij}(t)]$ is type k's *positive definite* covariance matrix. For convenience, and with some abuse of terminology, we refer to the matrix $\mathbf{V}^k(t)$ as the *inverse covariance matrix* of investor type k.

The equations in (5) show that type k's demand for asset j depends upon the prices prevailing in the other asset market; on their expectations for each asset given by their price beliefs $\mathbb{E}^k_t P_{t+1}$ and the elements $v^k_{ij}(t)$ of their inverse covariance matrix; and on the Lagrange multipliers $\lambda^k_{1,t}, \lambda^k_{2,t}$ on the short-selling constraints of the risky assets, $j = 1,2$.

3.2 Beliefs and population shares

We allow dynamic beliefs about prices and covariances as explained below. Further, the population shares of investor types are updated over time via *evolutionary competition*.

3.2.1 Beliefs

We assume that price beliefs and covariances are *boundedly-rational* and *predetermined* at date t (aside from a linear part, $\overline{c}P_t$, in the vector of price beliefs that may be set at zero).

Assumption 1 *Let $\overline{c} \in [0, 1+r)$, $L \geq 1$. All price beliefs and covariances have the form:*

$$\mathbb{E}^k_t P_{t+1} = \overline{c}P_t + \tilde{f}^k_t(P^f_t, P_{t-1}, ..., P_{t-L}, \mathbf{s}_t) \tag{6}$$

$$\mathbf{\Omega}^k(t) = \tilde{g}^k_t(\sigma^k(t-1), \sigma(t-1))\mathbf{\Omega}^k(t-1) \tag{7}$$

where $\tilde{f}^k_t \in \mathbb{R}^2$ is a forecast that cannot depend on date t prices, $P^f_t \in \mathbb{R}^2_+$ is a vector of fundamental asset prices, $\mathbf{s}_t$ is a vector of other factors that affect price beliefs, $\tilde{g}^k_t > 0$ is a scaling of k's past covariance, and $\sigma^k(t)$ ($\sigma(t)$) are vectors of the variances of assets 1,2.

Assumption 1 allows a range of boundedly-rational beliefs. Equation (6) allows linear dependence of price expectations on the current price via the coefficient $\overline{c}$; we assume $\overline{c} \geq 0$ to allow the case of no dependence on the current price (i.e. $\overline{c} = 0$) and we assume $\overline{c} < 1 + r$ to ensure the usual slope-sign is preserved; see (5). The case $\overline{c} = 1$, $\tilde{f}^k_t(.) = 0$ is a random walk model of asset prices. Extrapolation or 'trend-following' (e.g. Barberis et al., 2018) is permitted via inclusion of price lags up to a maximum lag L, and price expectations can depend on asset fundamentals via the fundamental price vector P^f_t, or on

other predetermined variables in vector $\mathbf{s}_t$.[3] The forecast $\tilde{f}_t^k(.)$ (which can differ across types and over time) allows a potentially non-linear response to *past* prices; it may also contain 'fixed effects' or be influenced by social networks (via $\mathbf{s}_t$). It does not pose any difficulty to include many exogenous or predetermined endogenous variables in $\tilde{f}_t^k(.)$; some examples are expected dividends or price beliefs of other investors (see Supplementary Appendix).

Equation (7) says that type k's subjective covariance matrix is updated by scaling by a coefficient $\tilde{g}_t^k(.) > 0$, which may depend on k's past variances $\sigma^k(t-1)$ and measures of the realized asset variances, $\sigma(t-1) = (\sigma_{11}(t-1), \sigma_{22}(t-1))'$ (e.g. from a sample of past returns). Because covariances $\sigma_{ij}^k(t)$ are scaled for each investor (see (7)), the covariance matrices $\mathbf{\Omega}^k(t)$ and the inverse matrices $\mathbf{V}^k(t)$ will remain positive definite under updating.

The scaling coefficient $\bar{c}$ in (6) is common and taken as fixed over time; however, it is easy to relax this assumption so that the scaling coefficients are *heterogeneous* or *time-varying*. We provide further details on these simple extensions in the Supplementary Appendix.

3.2.2 Population shares

We allow the population shares n_t^k following each forecasting rule to be endogenous and time-varying, but we rule out dependence on current and future prices $P_t, ..., P_T$ (Assumption 2).

Assumption 2 *We consider population shares $n_t^k = n_k(\mathbf{n}_{t-1}, \mathbf{u}_{t-1})$ where n_k is a function such that $n_t^k \in (0,1)$ $\forall t, k$, $\sum_{k\in[K]} n_t^k = 1$, and $\mathbf{n}_{t-1}$ ($\mathbf{u}_{t-1}$) is the vector of past population shares (resp. past fitness levels). In particular, we rule out dependence of n_t^k on current or future price vectors $P_t, ..., P_T$ but not dependence on past price vectors P_{t-1}, P_{t-2} etc.*

Assumption 2 is quite general. For example, the population shares can be determined by *evolutionary competition*. Following Brock and Hommes (1997), a popular approach is a discrete choice logistic model $n_{t+1}^k = \frac{\exp(\beta U_t^k)}{\sum_{k\in[K]} \exp(\beta U_t^k)}$, where the intensity of choice $\beta \in [0, \infty)$ determines how fast agents switch to better-performing predictors. Various fitness measures U_t^k are used in the literature, including realized profits (Brock and Hommes, 1998) and forecast accuracy (Ap Gwilym, 2010). Assumption 2 rules out the 'extreme' population shares of 0 or 1 and the population shares above are *not* asset-specific; however, both these simplifying assumptions can be relaxed quite easily, as shown in the Supplementary Appendix.

Having past fitness levels $\mathbf{u}_{t-1}$ in $n_k(.)$ permits evolutionary competition with extended memory, while the inclusion of past population shares $\mathbf{n}_{t-1}$ allows asynchronous updating (see e.g. Hommes, 2013). It does not impose any extra difficulty to allow the function n_k to be time-varying or to add further predetermined endogenous variables in the vector $\mathbf{u}_{t-1}$; however, as noted in Assumption 2, we rule out dependence of population shares on *current* or future prices. Fixed population shares $n_t^k = 1/K$ are relevant for agent-based or social network-based models where types are *individuals*, while herding models imply $n_t^k = h_t^k/K$, where h_t^k is the number of investors that follow rule k at date t (see e.g. Kirman, 1993).

[3]The fundamental price typically equals the *intrinsic value* of the asset (e.g. the expected discounted stream of future dividends); see, for example, Brock and Hommes (1998) and Section 5.1 below.

3.3 Market clearing

Using (6), the demands in (5) can be written succinctly as

$$N_t^k = \mathbf{V}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) + \mathbf{V}^k(t)\lambda_t^k, \qquad \text{for } k = 1, ..., K \tag{8}$$

with $N_t^k := (N_{1,t}^k, N_{2,t}^k)'$, $\lambda_t^k := (\lambda_{1,t}^k, \lambda_{2,t}^k)'$ and forecast and price vectors defined as

$$\mathbb{F}_t^k := (\mathbb{F}_{1,t}^k, \mathbb{F}_{2,t}^k)', \qquad \tilde{P}_t := (1 + \tilde{r})(P_{1,t}, P_{2,t})', \qquad \tilde{r} := r - \overline{c},$$

where $\mathbb{F}_{j,t}^k := \tilde{f}_{j,t}^k(.)$ for $j = 1, 2$. We now define a market equilibrium.

Definition 1 *A vector $\tilde{P}_t \in \mathbb{R}^2$ is an equilibrium price vector if there exists demands $N_t^1, ..., N_t^K \in \mathbb{R}_+^2$ and shadow prices $\lambda_t^1, ..., \lambda_t^K \in \mathbb{R}_+^2$ such that*

$$N_t^k = \mathbf{V}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) + \mathbf{V}^k(t)\lambda_t^k, \qquad \forall k \in [K] \tag{9}$$

$$(N_t^k)'\lambda_t^k = 0, \qquad \forall k \in [K] \tag{10}$$

$$\sum_{k=1}^{K} n_t^k N_t^k = \overline{N} \tag{11}$$

where both entries of $\overline{N} := (\overline{N}_1, \overline{N}_2)'$ are strictly positive.

To find equilibrium asset prices subject to (9)–(11), we use a guess-verify approach. Our starting point is the observation that the vector of Lagrange multipliers λ_t^k may be (i) zero (if investor k has all slack demands), (ii) strictly positive (if investor k is short-selling constrained for all assets), or (iii) have some zero elements and some positive elements. In all three cases, the demands can be written as $N_t^k = \hat{\mathbf{V}}^k(t)(\mathbb{F}_t^k - \tilde{P}_t)$ for some conformable matrix $\hat{\mathbf{V}}^k(t)$. By guessing on the structure of this matrix and finding conditions such that the guess is verified, we can relate equilibrium prices to investor beliefs, as shown below.

Finding an equilibrium is non-trivial in practice due to the *curse of dimensionality*. To see the problem, note the number of cases (without pruning) is 4^K, implying rapid combinatorial explosion. For $K = 10$, there are around 1 million cases, and for $K = 20$, there are more than 1 *trillion* cases, so even modestly-sized problems pose a severe computational burden. In the analysis that follows, we show how to tame this curse of dimensionality.

3.4 Equilibrium prices and demands

We now characterize equilibrium prices and demands. By Theorem 2 in Tong et al. (2017b), there exists a unique equilibrium price vector that satisfies Definition 1 for given beliefs and covariance matrices. Hence, there is a unique equilibrium price vector at each date $t \in [T]$.[4]

[4]Theorem 2 in Tong et al. (2017b) states there exists an equilibrium price vector $\tilde{P}^*$ that is unique if the outside supplies for each asset ($j = 1, ..., J$) differ from zero. The case $J = 2$ and $\overline{N}_1, \overline{N}_2 > 0$ here is nested.

It is easy to show that investor k's demands for assets $i = 1, 2$ are:[5]

$$N_{i,t}^k = \begin{cases} \mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) & \text{if } \mathbf{V}_1^k(t)(\mathbb{F}_t^k - \tilde{P}_t), \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0, \quad (*) \\ \frac{|\mathbf{V}^k(t)|}{v_{-i-i}^k(t)}(\mathbb{F}_{i,t}^k - \tilde{P}_{i,t}) & \text{if } \mathbb{F}_{i,t}^k \geq \tilde{P}_{i,t},\ \mathbf{V}_{-i}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) < 0, \quad (**) \\ 0 & \text{if } \mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) < 0,\ \mathbb{F}_{-i,t}^k \geq \tilde{P}_{-i,t} \text{ or } \tilde{P}_{i,t} > \mathbb{F}_{i,t}^k\ \forall i \ (***) \end{cases} \tag{12}$$

where $|\mathbf{V}^k(t)| = \det(\mathbf{V}^k(t))$ and $-i$ is the other asset (i.e. the only member of $\{1, 2\} \setminus \{i\}$).

The above demands can be written succinctly as

$$N_{i,t}^k = \hat{\mathbf{V}}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t), \qquad i = 1, 2 \tag{13}$$

where $\hat{\mathbf{V}}_i^k(t) = \mathbf{V}_i^k(t)$ if (*) holds, $\hat{\mathbf{V}}_i^k(t) = \mathbf{0}$ if (***) holds, and $\hat{\mathbf{V}}_i^k(t) = \frac{|\mathbf{V}^k(t)|}{v_{-i-i}^k(t)}[\mathbb{1}_{i=1} \ \ \mathbb{1}_{i=2}]$ if (**) holds, where $\mathbb{1}_{i=x}$ is an indicator variable equal to 1 if $i = x$ and 0 otherwise.

To track whether each investor belongs to case (*), (**) or (***) in (12), we let $B_{12,t}$ be the set of 'buyers' of both assets at date t and let $B_{11,t}, B_{22,t}$ be the sets of investors in asset 1 only, resp. asset 2 only. These sets are disjoint and may be empty, but $B_{1,t} := B_{11,t} \cup B_{12_t}$ and $B_{2,t} := B_{22,t} \cup B_{12,t}$ must be non-empty at equilibrium (each asset must be held by *some* type). Similarly, let $S_{i,t}$ be the set of investors who are short-selling constrained for asset i at date t, so $S_{12,t} := S_{1,t} \cap S_{2,t}$ is the set of short-selling constrained types for both assets.

By (12), there are four possible combinations of demands for each type: (slack, slack), (slack, constrained), (constrained, slack), (constrained, constrained). As noted, there is a unique equilibrium price vector. We now state a proposition that draws on the above results.

Proposition 1 *Let $\tilde{\mathbb{F}}_t, \hat{\mathbb{F}}_t, \tilde{N}_t, \hat{N}_t$ be the vectors defined below and $n_t^k \in (0, 1)\ \forall k$. Let $B_{i,t} \subseteq [K]$ be a non-empty set of 'buyers' of asset $i \in \{1, 2\}$; $S_{i,t} := [K] \setminus B_{i,t}$ contains all other types. Let $B_{ii,t}$ contain buyers of asset i only, and let $B_{12,t} := B_{1,t} \cap B_{2,t}$, $S_{12,t} := S_{1,t} \cap S_{2,t}$.*

(i) If $\min_{k \in [K]} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t) \right) \geq 0$ for $i = 1, 2$, short-selling constraints are slack for all investors ($B_{1,t} = B_{2,t} = [K]$) and the equilibrium price vector and demands at date t are

$$\tilde{P}_t = \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t) \right]^{-1} \left(\sum_{k=1}^K n_t^k \mathbf{V}^k(t) \mathbb{F}_t^k - \overline{N} \right) \tag{14}$$

$$N_{i,t}^k = \mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \quad \forall k \in [K],\ i = 1, 2 \tag{15}$$

where $\tilde{\mathbb{F}}_t = \sum_{k=1}^K \overline{\mathbf{V}}^k(t) \mathbb{F}_t^k$, $\overline{\mathbf{V}}^k(t) = \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t) \right]^{-1} n_t^k \mathbf{V}^k(t)$, $\tilde{N}_t = \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t) \right]^{-1} \overline{N}$.

(ii) If $\min_{k \in [K]} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t) \right) < 0$ for some $i \in \{1, 2\}$, then at least one investor type is short-selling constrained for some risky asset and there exists sets $B_{12,t}, S_{12,t}, B_{ii,t} \subset [K]$

[5] See Supplementary Appendix. For reference, we also consider there the benchmark of one risky asset.

such that the following conditions hold for $i = 1, 2$*:*

$$\min_{k \in B_{12,t}} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t) \right) \geq 0, \quad \max_{k \in S_{12,t}} \left(\mathbb{F}_{i,t}^k \right) < \hat{\mathbb{F}}_{i,t} - \hat{N}_{i,t} \tag{16}$$

$$\min_{k \in B_{ii,t}} \left(\mathbb{F}_{i,t}^k \right) \geq \hat{\mathbb{F}}_{i,t} - \hat{N}_{i,t}, \quad \max_{k \in B_{ii,t}} \left(\mathbf{V}_{-i}^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t) \right) < 0 \tag{17}$$

where $\hat{N}_t = \left[\sum_{k \in B_{1,t} \cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t) \right]^{-1} \overline{N}$*, and the equilibrium price vector and demands are*

$$\tilde{P}_t = \left[\sum\nolimits_{k \in B_{1,t} \cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t) \right]^{-1} \left(\sum_{k \in B_{1,t} \cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t) \mathbb{F}_t^k - \overline{N} \right) \tag{18}$$

$$N_{i,t}^k = \hat{\mathbf{V}}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \ \ \forall k \in B_{i,t}, \quad N_{i,t}^k = 0 \ \ \forall k \in S_{i,t}, \quad i = 1, 2 \tag{19}$$

where $\hat{\mathbf{V}}^k(t) = \mathbf{V}^k(t)$ *if* $k \in B_{12,t}$*,* $\hat{\mathbf{V}}^k(t)$ *has one non-zero entry* $\hat{v}_{ii}^k(t) = \frac{|\mathbf{V}^k(t)|}{v_{-i-i}^k(t)}$ *if* $k \in B_{ii,t}$*, and* $\hat{\mathbb{F}}_t = \sum_{k \in B_{1,t} \cup B_{2,t}} \overline{\mathbf{V}}^k(t) \mathbb{F}_t^k$*, with* $\overline{\mathbf{V}}^k(t) = \left[\sum_{k \in B_{1,t} \cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t) \right]^{-1} n_t^k \hat{\mathbf{V}}^k(t)$*.*

Proof. See the Appendix. ■

Proposition 1 gives active-slack conditions on beliefs $\{\mathbb{F}_t^k, \mathbf{V}^k(t)\}_{k=1}^K$. Part (i) provides conditions on beliefs such that all short-selling constraints are *slack* in equilibrium. In this case, prices and demands follow the standard formulas when positions are unrestricted; thus, if risk-weighted belief dispersion is small enough (i.e. $\min_{k \in [K]} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t) \right) \geq 0$) the outcome is the same as if short-selling constraints were absent in period t.

If beliefs do not satisfy the slackness condition in Part (i), some short-selling constraints are *binding* and we are in Part (ii) of Proposition 1. Any constrained investors are collected in the sets $S_{1,t}$ and $S_{2,t}$, whose union must be non-empty; and the 'buyers' of each asset are collected in the non-empty sets $B_{1,t}, B_{2,t}$, where $B_{i,t}$ contains both investors who buy both assets ($B_{12,t}$) plus investors who buy only asset i ($B_{ii,t}$). The sets $S_{i,t}$ and $B_{i,t}$ are determined by cut-off conditions on beliefs (see (16)–(17)) and the equilibrium price vector in (18) depends on the sum taken over the set of buyers of one or more assets $B_{1,t} \cup B_{2,t}$, i.e. the market participants.[6] Note from (19) that the members of different sets of investors (holders of one versus both versus no assets) differ in the matrices $\hat{\mathbf{V}}^k(t)$, as in (12)–(13).

The buyers of an asset are those who are sufficiently optimistic about its risk-weighted future value, given their subjective covariance matrix and risk aversion. The non-participants are 'pessimists' who are priced out of the market (see (16)–(19)). The price vector $\tilde{P}_t$ is unique as already noted. The total number of cases in Proposition 1 corresponds to the different combinations for the sets of buyers and constrained investors. There are four cases for the

[6] If a set is empty, the relevant condition in (16) or (17) is vacuously true (the 'if' part of "if in set, then..." is false). Note: if $B_{12,t} = \emptyset$ then $B_{11,t}, B_{22,t} \neq \emptyset$, and if $k \in S_{12,t}$, then $k \notin B_{i,t} = B_{12,t} \cup B_{ii,t}$, $i \in \{1, 2\}$.

demands of each investor type; see (12).[7] Given K investors, this makes 4^K cases in each period, t, including some cases that are infeasible (no buyers of some asset). Subtracting out the infeasible cases gives the total number of cases as $4^K-(2^{K+1}-1)$. Finding the equilibrium at date t amounts to determining the sets in Proposition 1 to which each investor belongs.

Example 1 *Consider the static example in Jarrow (1980) of two investors with identical risk aversion coefficients of 1 (i.e. $a^1 = a^2 = 1$; $T = 1$, $n^1 = n^2 = 1/2$). The first investor prefers asset 1, the second investor prefers asset 2, and each asset has outside supply of 1/2:*

$$\mathbb{F}^1 = \mathbb{E}^1 P = (2, 1), \quad \mathbb{F}^2 = \mathbb{E}^2 P = (1, 3), \qquad \overline{N} = (1/2, 1/2)$$

$$\mathbf{\Omega}^1 = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{\Omega}^2 = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \quad \Longrightarrow \quad \mathbf{V}^1 = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}, \quad \mathbf{V}^2 = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix}.$$

We can use Proposition 1 to find the equilibrium price vector. There is no equilibrium where both investors buy both assets because the guessed equilibrium prices and demands are

$$\tilde{P} = (7/6,\ 11/6), \qquad N^1 = (10/6,\ -5/6), \qquad N^2 = (-4/6,\ 11/6)$$

such that the short-selling constraints $N_1^k, N_2^k \geq 0$ are violated for both assets.

Let us guess that Investor 1 buys only asset 1 and Investor 2 buys only asset 2, such that $B_{12} = \emptyset$, $B_1 = B_{11} = \{1\}$, $B_2 = B_{22} = \{2\}$, and $S_1 = \{2\}$, $S_2 = \{1\}$.

Given this guess and $\hat{v}_{ii}^k = \frac{|\mathbf{V}^k|}{v_{-i-i}^k}$ if $k \in B_{ii}$ and 0 otherwise,

$$\hat{\mathbf{V}}^1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \hat{\mathbf{V}}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

and hence, by Proposition 1, the price vector and demands are

$$\tilde{P} = \left[\sum_{k=1}^{2} n^k \hat{\mathbf{V}}^k\right]^{-1} \left(\sum_{k=1}^{2} n^k \hat{\mathbf{V}}^k \mathbb{F}^k - \overline{N}\right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (= \hat{\mathbb{F}} - \hat{N})$$

$$N^1 = \hat{\mathbf{V}}^1(\mathbb{F}^1 - \tilde{P}) = (1,\ 0), \quad N^2 = \hat{\mathbf{V}}^2(\mathbb{F}^2 - \tilde{P}) = (0,\ 1)$$

and this is an equilibrium because $\sum_{k=1}^{2} n^k N^k = (1/2, 1/2)$ and

$$\mathbf{V}_2^1(\mathbb{F}^1 - \tilde{P}) = -1 < 0, \qquad \mathbf{V}_1^2(\mathbb{F}^2 - \tilde{P}) = -0.5 < 0, \qquad \mathbb{F}_i^{k=i} > \tilde{P}_i, \ \ i = 1, 2.$$

The price of asset 2 rises relative to unrestricted short-selling, but the price of asset 1 falls. Hence, the price increase result of Miller (1977) does not extend to a multi-asset setting, as shown by Jarrow (1980). We use the above example to illustrate Proposition 1. It is easy

[7] The four cases are (slack, slack), (slack, constrained), (constrained, slack), (constrained, constrained).

but tedious to confirm that $\tilde{P} = (1, 2)$*,* $N^1 = (1, 0)$*,* $N^2 = (0, 1)$ *is the only equilibrium; note that there are 9 cases in total (including the slack case we rejected above).*

The general case of K investors and 2 assets is intractable given $4^K - (2^{K+1} - 1)$ candidates for the sets $B_{11,t}, B_{22,t}, B_{12,t}, S_t$. For instance, with $K = 3$ investors there are 49 cases to consider, for $K = 5$ investors there are 961 cases, and for $K = 10$ there is more than 1 million cases ($\approx 1.05 \times 10^6$). Clearly, even these small-dimension examples are analytically intractable, and increasing K soon leads to computationally intractable examples, as cases increase exponentially (e.g. for $K = 30$, there are more than 1 billion billion cases!).

Given this curse of dimensionality, we now present some economically-interesting cases where the problem remains tractable even if the number of investors is large. These cases include investors with optimism aligned across assets, and a selective short-selling ban.

4 Taming the curse of dimensionality

4.1 Short-selling ban on one asset

Consider a short-selling ban on one asset only. This case has some empirical realism because, in practice, short-selling bans have often been introduced by regulators specifically for certain types of risky assets, such as the shares of particular industry, as opposed to a blanket short-selling ban on all shares in the market (e.g. the selective ban on shorting UK financial stocks in 2008–9, or the SEC's Emergency Order in 2008; see Siciliano and Ventoruzzo (2020)).

With a short-selling constraint on asset 2 only, type k's demands are:[8]

$$N_{1,t}^k = \begin{cases} \mathbf{V}_1^k(t)(\mathbb{F}_t^k - \tilde{P}_t) & \text{if } \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \\ \frac{|\mathbf{V}^k(t)|}{v_{22}^k(t)}(\mathbb{F}_{1,t}^k - \tilde{P}_{1,t}) & \text{if } \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) < 0 \end{cases} \tag{20}$$

$$N_{2,t}^k = \begin{cases} \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) & \text{if } \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \\ 0 & \text{if } \mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{P}_t) < 0. \end{cases} \tag{21}$$

Equations (20)–(21) show that investors always take a position in asset 1 (either long or short, depending on their expectations), while for asset 2 they participate if they are not short-selling constrained. Note that investor k's position in asset 1 depends on whether they take a position in asset 2 or are short-selling constrained; compare (20) and (21).

The demands can be derived by setting $\lambda_{1,t}^k = 0 \; \forall k$ in the first-order conditions, (8). The result is just two cases per investor (buy asset 2 or take a constrained position of zero), as shown in Figure 1 below. Given two cases per investor and K investors, the total number of cases is 2^K; however, one of these cases (no investor holds asset 2) cannot be an equilibrium given positive outside supplies; see Figure 1, bottom right. Thus, the relevant number of

[8] For further details, see the third section of the Supplementary Appendix (available here).

cases is $2^K - 1$, i.e. exponential in K with base 2 – a substantial improvement. We note this result in Remark 1. A formal statement appears in the Supplementary Appendix.

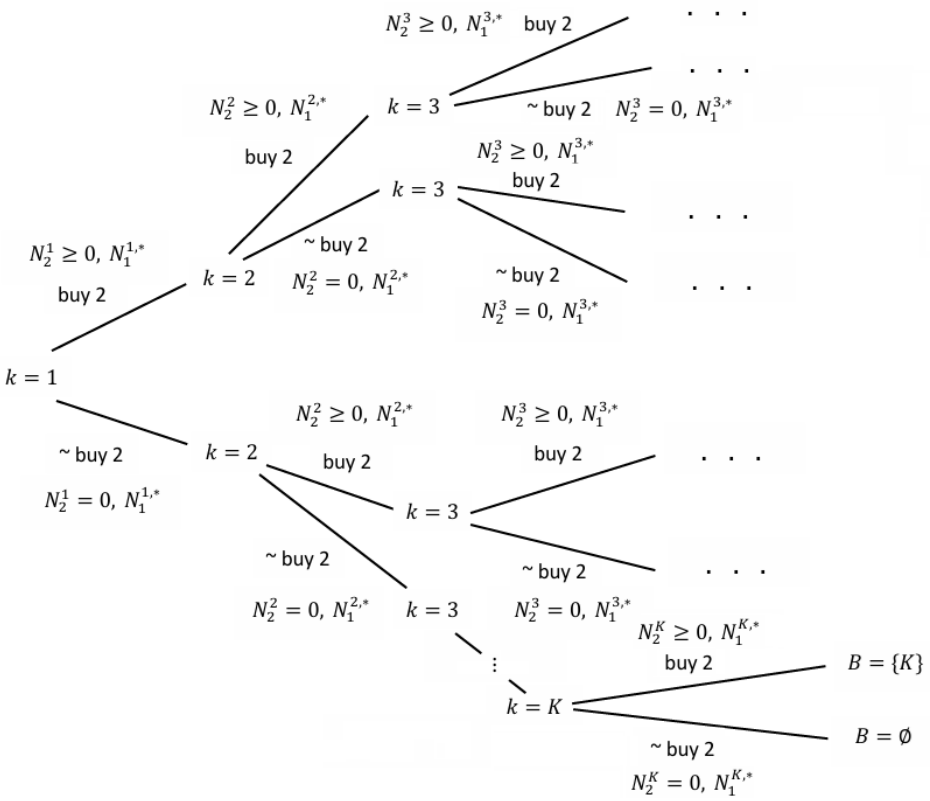


Figure 1: Tree diagram of the $2^K - 1$ cases: short-selling constraint on asset 2 only. There are two cases for each investor, so the number of branches (cases) doubles with each node (investor). The -1 correction arises because one of the 2^K terminal branches corresponds to the outcome that there is no buyer of asset 2 (see bottom right). Note that buy 2 ($\sim$ buy 2) means a slack (binding) short-selling constraint for asset 2. Time subscripts are omitted.

Remark 1 *With a short-selling constraint on Asset 2 only, the results in Proposition 1 are simplified. All slack constraints holds if* $\min_{k\in[K]}\left(\mathbf{V}_2^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t)\right) \geq 0$*; then the equilibrium price vector* $\tilde{P}_t$ *follows Proposition 1 Part (i) and* $N_{1,t}^k \in \mathbb{R}$, $N_{2,t}^k \geq 0\ \forall k \in [K]$. *Otherwise, a short-selling constraint binds for at least one type and there exists sets* $B_t, S_t \neq \emptyset$ *such that* $\max_{k\in S_t}\left(\mathbf{V}_2^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t)\right) < 0 \leq \min_{k\in B_t}\left(\mathbf{V}_2^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t)\right)$ *and* $\tilde{P}_t$ *follows Proposition 1 Part (ii), except the sum is taken over all types because short-selling of Asset 1 is allowed, and* $N_{1,t}^k \in \mathbb{R}\ \forall k \in [K]$, $N_{2,t}^k \geq 0\ \forall k \in B_t$, $N_{2,t}^k = 0\ \forall k \in S_t\ (= [K] \setminus B_t)$.

There are far fewer cases than for Proposition 1. Intuitively, investors can be partitioned into just two sets, B_t and S_t, as there are only *two* cases for the demands for each asset, in

contrast to *four* in the double-ban case; see (12) and (20). The total number of cases falls from $4^K - (2^{K+1} - 1)$ to $2^K - 1$, a drastic improvement since the leading term falls from 4^K to 2^K. We exploit this result below by showing that when beliefs are *ordered by optimism*, the number of cases falls to K (linear), which is highly computationally tractable.

4.1.1 Ranking investor beliefs

We assume investor types have common inverse covariance matrices $\mathbf{V}^k(t) = \overline{\mathbf{V}}(t)\ \forall k$ (we discuss below how this assumption can be relaxed) and that beliefs can be ordered as

$$\overline{\mathbf{V}}_2(t)\mathbb{F}_t^1 < \overline{\mathbf{V}}_2(t)\mathbb{F}_t^2 < \ldots < \overline{\mathbf{V}}_2(t)\mathbb{F}_t^K \tag{22}$$

where $\overline{\mathbf{V}}_2(t)$ is the second row of the common (inverse) covariance matrix at date t.

Inequality (22) orders investors according to their risk-adjusted optimism for asset 2 (more optimistic types receive a higher label, k). Although (22) rules out ties in terms of optimism, such ties may be broken through minor changes in price beliefs $\mathbb{F}_t^k$ or the common covariance matrix. Finally, note that (22) is simply a re-labelling of investor identities that can be made for any set of beliefs $\{\overline{\mathbf{V}}(t), \mathbb{F}_t^1, \ldots, \mathbb{F}_t^K\}$ that satisfy the inequality.

By (22), we have the following result that greatly simplifies the analytics and makes the task of finding a numerical solution computationally tractable (K cases: linear).

Corollary 1 *Suppose inequality* (22) *holds and only asset 2 is subject to a short-selling constraint. If* $\overline{\mathbf{V}}_2(t)\left(\sum_{k=1}^K n_t^k \mathbb{F}_t^k - \mathbb{F}_t^1\right) \leq \overline{\mathbf{V}}_2(t)[\overline{\mathbf{V}}(t)]^{-1}\overline{N}$*, then all investors types* $k \in [K]$ *are unconstrained and the equilibrium price vector and demands are*

$$\tilde{P}_t = \sum_{k\in[K]} n_t^k \mathbb{F}_t^k - [\overline{\mathbf{V}}(t)]^{-1}\overline{N} := \tilde{P}_{(0)}$$

$$\forall k \in [K]: \quad N_{1,t}^k = \overline{\mathbf{V}}_1(t)(\mathbb{F}_t^k - \tilde{P}_t) \in \mathbb{R}, \quad N_{2,t}^k \geq 0.$$

Otherwise, $\exists$ *a unique* $\tilde{k} \in [K-1]$ *s.t.* $\overline{\mathbf{V}}_2(t)(\hat{\mathbb{F}}_{t,(\tilde{k})} - \mathbb{F}_t^{\tilde{k}+1}) \leq \overline{\mathbf{V}}_2(t)\hat{N}_{t,(\tilde{k})} < \overline{\mathbf{V}}_2(t)(\hat{\mathbb{F}}_{t,(\tilde{k})} - \mathbb{F}_t^{\tilde{k}})$ (*where vectors* $\hat{\mathbb{F}}_{t,(\tilde{k})}$, $\hat{N}_{t,(\tilde{k})}$ *are defined below*)*, investor types* $1, ..., \tilde{k}$ *are constrained, types* $\tilde{k}+1, ..., K$ *are unconstrained, and the equilibrium price vector and demands are*

$$\tilde{P}_t = \hat{\mathbb{F}}_{t,(\tilde{k})} - \hat{N}_{t,(\tilde{k})} := \tilde{P}_{t,(\tilde{k})}$$

$$N_{1,t}^k = \overline{\mathbf{V}}_1(t)(\mathbb{F}_t^k - \tilde{P}_t)\ \forall k, \quad N_{2,t}^k = 0\ \ \forall k \leq \tilde{k}, \quad N_{2,t}^k = \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0\ \ \forall k > \tilde{k}$$

where $\hat{\mathbb{F}}_{t,(\tilde{k})} = [\sum_{k=1}^{\tilde{k}} n_t^k \hat{\mathbf{V}}(t) + \sum_{k=\tilde{k}+1}^K n_t^k \overline{\mathbf{V}}(t)]^{-1}\left(\sum_{k=1}^{\tilde{k}} n_t^k \hat{\mathbf{V}}(t)\mathbb{F}_t^k + \sum_{k=\tilde{k}+1}^K n_t^k \overline{\mathbf{V}}(t)\mathbb{F}_t^k\right)$, $\hat{N}_{t,(\tilde{k})} = [\sum_{k=1}^{\tilde{k}} n_t^k \hat{\mathbf{V}}(t) + \sum_{k=\tilde{k}+1}^K n_t^k \overline{\mathbf{V}}(t)]^{-1}\overline{N}$, $\hat{\mathbf{V}}(t)$ *has one non-zero entry* $\hat{v}_{11}(t) = \frac{|\overline{\mathbf{V}}(t)|}{\overline{v}_{22}(t)}$, *and* $\tilde{P}_{t,(\tilde{k})}$ *is the equilibrium price vector when types* $1, ..., \tilde{k}$ *are short-selling constrained.*

Proof. It follows from Remark 1 and its proof in the Supplementary Appendix. ■

By Corollary 1, there are K cases for the equilibrium price vector and demands (in a given period) when investor types are ordered in terms of optimism as in (22). As such, the number of cases scales *linearly* with the number of investor types K, making it computationally tractable to a find a solution even for very large K, i.e. thousands or even millions of investors.

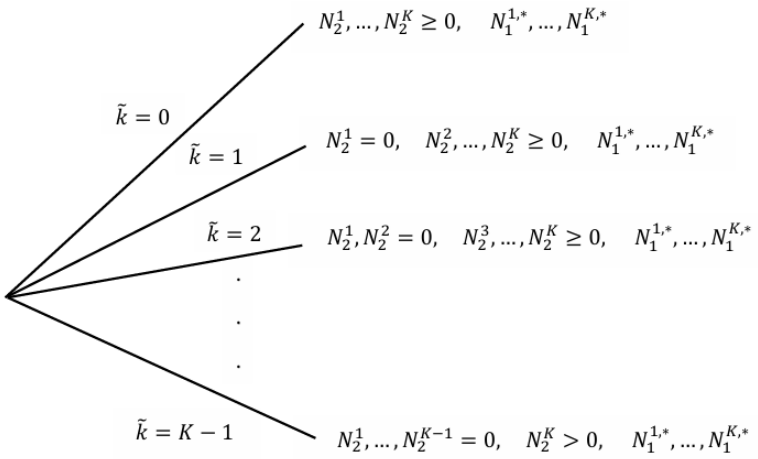


Figure 2: Tree diagram of the K cases when types are sorted as 1 (most pessimistic) up to K (most optimistic). Each branch has $\tilde{k}$ short-selling constrained types for asset 2, i.e. types $1, ..., \tilde{k}$ are constrained and types $\tilde{k}+1, ..., K$ are not. The unconstrained demands for asset 1 are represented as $N_1^{k,*}$ for $k = 1, ..., K$. Time subscripts are omitted to reduce clutter.

The intuition for K cases is shown in Figure 2. Each branch corresponds to a different number of short sellers $\tilde{k}$, with $\tilde{k} = 0$ being the special case where all types have slack constraints and $\tilde{k} = K - 1$ being the case where only type K (the most optimistic) buys asset 2 (i.e. constraints are active for types $1, ..., K-1$). Note that $N_1^{1,*}, ..., N_1^{K,*}$ are the unrestricted demands for asset 1. At a given branch with $\tilde{k} > 1$ we see that types $1, ..., \tilde{k}$ are constrained because investor $\tilde{k}$ is sufficiently pessimistic (by definition) to be a constrained type, and $1, ..., \tilde{k}-1$ are more pessimistic than investor $\tilde{k}$, so they must also be short-selling constrained. Similarly, all types $k \geq \tilde{k}+1$ will be 'buyers' of asset 2 because the marginal investor $\tilde{k}+1$ is unconstrained and is the least optimistic in the group of 'optimists'. As a result, the number of cases to be considered is reduced from $2^K - 1$ to K (many infeasible candidates for equilibrium sets have been excluded). We now provide a simple example.

Example 2 *Consider again the static Example 1. We assume both types ($n^1 = n^2 = 1/2$) have identical covariance matrices and there is a short-selling constraint on asset 2 only:*

$$\mathbb{F}^1 = (2, 1), \quad \mathbb{F}^2 = (1, 3), \quad \overline{N} = (1/2, 1/2)$$

$$\mathbf{\Omega}^1 = \mathbf{\Omega}^2 = \overline{\mathbf{\Omega}} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \quad \Longrightarrow \quad \overline{\mathbf{V}} = \overline{\mathbf{\Omega}}^{-1} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}.$$

Note that $\overline{\mathbf{V}}_2\mathbb{F}^2 = 1 > \overline{\mathbf{V}}_2\mathbb{F}^1 = -1/2$, *so* (22) *holds (without any relabelling). By Corollary 1, there are only* $K = 2$ *candidates for the equilibrium price vector:*

$$\tilde{P} = \begin{cases} \frac{1}{2}\left(\mathbb{F}^1 + \mathbb{F}^2\right) - \overline{\mathbf{V}}^{-1}\overline{N} & \text{if } \frac{1}{2}\overline{\mathbf{V}}_2(\mathbb{F}^2 - \mathbb{F}^1) \leq \overline{\mathbf{V}}_2\overline{\mathbf{V}}^{-1}\overline{N} \\ \hat{\mathbb{F}}_{(1)} - \hat{N}_{(1)} & \text{if } \overline{\mathbf{V}}_2(\hat{\mathbb{F}}_{(1)} - \mathbb{F}^1) > \overline{\mathbf{V}}_2\hat{N}_{(1)} \end{cases}$$

where $\hat{N}_{(1)} = (\hat{\mathbf{V}} + \overline{\mathbf{V}})^{-1}2\overline{N} = (1, 3)$, $\hat{\mathbf{V}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\hat{\mathbb{F}}_{(1)} = (\hat{\mathbf{V}} + \overline{\mathbf{V}})^{-1}\left(\hat{\mathbf{V}}\mathbb{F}^1 + \overline{\mathbf{V}}\mathbb{F}^2\right) = \left(\frac{3}{2}, \frac{7}{2}\right)$.

The demands of investor types 1 and 2 are

$$N^1, N^2 = \begin{cases} \overline{\mathbf{V}}(\mathbb{F}^1 - \tilde{P}),\ \overline{\mathbf{V}}(\mathbb{F}^2 - \tilde{P}) & \text{if } \overline{\mathbf{V}}_2(\mathbb{F}^2 - \mathbb{F}^1) \leq \overline{\mathbf{V}}_2\overline{\mathbf{V}}^{-1}\overline{N} \\ \hat{\mathbf{V}}(\mathbb{F}^1 - \tilde{P}),\ \overline{\mathbf{V}}(\mathbb{F}^2 - \tilde{P}) & \text{if } \overline{\mathbf{V}}_2(\overline{\mathbb{F}}_{(1)} - \mathbb{F}^1) > \overline{\mathbf{V}}_2\hat{N}_{(1)}. \end{cases}$$

We first check for slackness. Since $\frac{1}{2}\overline{\mathbf{V}}_2(\mathbb{F}^2 - \mathbb{F}^1) = 3/4 > \overline{\mathbf{V}}_2\overline{\mathbf{V}}^{-1}\overline{N} = 1/2$, *the guess of slackness is rejected. The remaining case is: (investor 1 constrained, investor 2 not), with*

$$\tilde{P} = \hat{\mathbb{F}}_{(1)} - \hat{N}_{(1)} = (1/2, 1/2), \qquad \overline{\mathbf{V}}_2(\hat{\mathbb{F}}_{(1)} - \mathbb{F}^1) = 13/4 > \overline{\mathbf{V}}_2\hat{N}_{(1)} = 1$$

and the corresponding demands are

$$N^1 = \hat{\mathbf{V}}(\mathbb{F}^1 - \tilde{P}) = (3/2,\ 0), \qquad N^2 = \overline{\mathbf{V}}(\mathbb{F}^2 - \tilde{P}) = (-1/2,\ 1).$$

It is easy to show that the price of asset 2 increases relative to unrestricted short-selling of both assets, while the price of asset 1 is unchanged. The relevant 'if' condition is satisfied.[9] *Due to beliefs being ordered, we only had to check only 2 cases to find the solution, as compared to the* $4^K - (2^{K+1} - 1) = 9$ *cases when types are not ordered by optimism*

4.1.2 Relaxing the covariance restriction

We assumed in Section 4.1.1 that investors have *identical* inverse covariance matrices, but we can relax this assumption. One simple case where the restriction can be relaxed is inverse covariance matrices $\mathbf{V}^k(t)$ with *identical second rows*, i.e. $\mathbf{V}^i(t) \neq \mathbf{V}^j(t)$ for $i \neq j$ but $\mathbf{V}_2^k(t) = \overline{\mathbf{V}}_2(t)\ \forall k \in [K]$. In this case, it is clear the inequality in (22) holds even for investor-specific covariance matrices, since only the second row enters. This case allows investor types to differ in their views about the riskiness of assets 1 and 2, via different covariance matrices, $\mathbf{\Omega}^k(t)$; Corollary 1 then changes slightly (see Supp. Appendix, Sec. 5).

Alternatively, suppose that investors differ in risk aversion coefficients or the scaling of their covariance matrices $\mathbf{\Omega}^k(t)$, such that $\mathbf{V}^k(t) = c^k\overline{\mathbf{V}}(t)$, where $c^k > 0\ \forall k$ and $\overline{\mathbf{V}}(t)$ is the

[9] Note that $\overline{\mathbf{V}}_2(\hat{\mathbb{F}}_{(1)} - \mathbb{F}^1) = 3/2 > \overline{\mathbf{V}}_2\hat{N}_{(1)} = 1$. Equivalently, $\overline{\mathbf{V}}_2(\mathbb{F}^1 - \tilde{P}) = -1/2 < 0$, so $\lambda_2 > 0$.

matrix of investor 1, with normalization $c^1 = 1$. Starting from (22), note that

$$\overline{\mathbf{V}}_2(t)\mathbb{F}_t^1 < \ldots < \overline{\mathbf{V}}_2(t)\mathbb{F}_t^K \quad \Longrightarrow \quad \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^1 - \tilde{P}_t) < \ldots < \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^K - \tilde{P}_t) \tag{23}$$

for any price vector $\tilde{P}_t$. For any equilibrium price vector such that short-selling constraints bind for some investor(s), there must exist a marginal investor $\tilde{k} \in [K-1]$ such that $\overline{\mathbf{V}}_2(t)(\mathbb{F}_t^1 - \tilde{P}_t) < \ldots < \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^{\tilde{k}} - \tilde{P}_t) < 0 \leq \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^{\tilde{k}+1} - \tilde{P}_t) < \ldots < \overline{\mathbf{V}}_2(t)(\mathbb{F}_t^K - \tilde{P}_t)$ (some type must hold asset 2 in equilibrium). By $sgn\left(c^k\overline{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t)\right) = sgn\left(\overline{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t)\right)$, we then get a scaling-amended version of the previous inequality:

$$\max_{k\in\{1,\ldots,\hat{k}\}}\left(c^k\overline{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t)\right) < 0 \leq \min_{k\in\{\hat{k}+1,\ldots,K\}}\left(c^k\overline{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t)\right) \tag{24}$$

where $\hat{k}$ may differ to $\tilde{k}$ since the price vector will generally change. The inequality in (24) shows that ranking investors as in (22) will still identify sets of unconstrained and constrained investors for asset 2, despite heterogeneous covariance matrices $\mathbf{V}^k(t) = c^k\overline{\mathbf{V}}(t)$.[10]

4.2 Aligned optimism

We now consider *aligned optimism*, which imposes consistency on the optimism ranking across assets. In particular, the ranking in terms of beliefs (both raw and risk-weighted) is either the same for both risky assets, or inverted for the second asset. We refer to an identical ranking for both assets as positive alignment, and that of inversion as negative alignment.

4.2.1 Positive alignment

Consider the benchmark of identical inverse covariance matrices: $\mathbf{V}^k(t) = \overline{\mathbf{V}}(t)$, $\forall k$. The positive alignment condition can then be expressed as follows:

$$\mathbb{F}_{i,t}^1 < ... < \mathbb{F}_{i,t}^K, \qquad \overline{\mathbf{V}}_i(t)\mathbb{F}_t^1 < \ldots < \overline{\mathbf{V}}_i(t)\mathbb{F}_t^K, \qquad i = 1, 2 \tag{25}$$

where investor types $k = 1, \ldots, K$ are ordered according to their optimism.

Intuitively, (25) says that an investor's relative ranking in terms of price beliefs is the same for both assets, and this ranking is preserved under 'risk-adjustment' using the inverse covariance matrix $\overline{\mathbf{V}}(t)$. By sorting, those investors with higher labels, k, are more optimistic. An example of positive alignment is trend-followers who use the same extrapolation coefficient (heuristic) for both assets that are over-valued relative to fundamentals (those with higher coefficients are more optimistic); note this optimism will be preserved under risk adjustment if the $\overline{\mathbf{V}}(t)$ matrix has all positive entries or if the off-diagonal entries are

[10] If investors update differently from a common initial covariance matrix, then the c^k coefficients become time-varying but are predetermined; note that there is no extra difficulty in this case (see (7)).

negative but sufficiently small in absolute value. We give an example of positive alignment in Section 5 using a two-asset Brock and Hommes (1998) model with short-selling constraints.

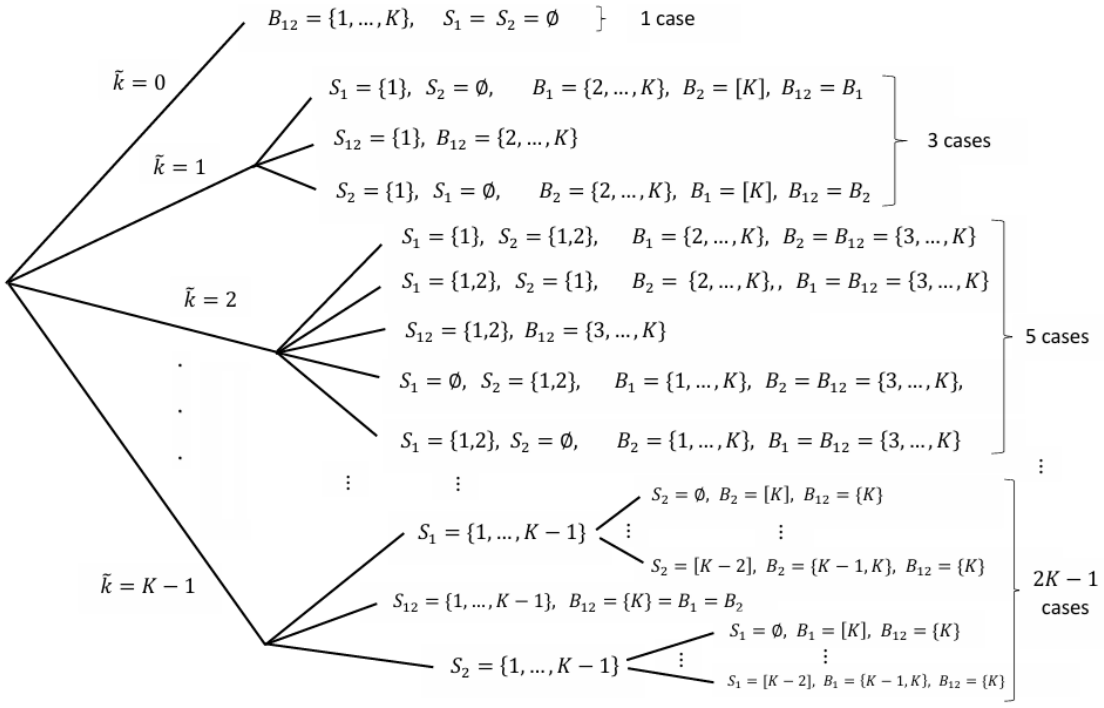


Figure 3: Tree diagram showing the cases when types are sorted as 1 (most pessimistic) up to K (most optimistic) and there is a short-selling ban on both assets: positive alignment. Each initial branch corresponds to $\tilde{k}$ investors who are short-selling constrained for some asset, and cases are shown by the terminal branches. Total no. of cases is $1+3+5+...+2K-1 = K^2$.

The ranking in (25) rules out many cases: an investor with ranking k' cannot be short-selling constrained if some type $k < k'$ is not. Similarly, if investor k' is short-selling constrained, then all types $k < k'$ (more pessimistic) must also be constrained. As a result, the number of cases is reduced to K^2 (i.e. exponential with base 4 to quadratic); see Figure 3.

We first consider whether a short-selling constraint will bind for some risky asset. We have said that if some investor is short-selling constrained for a particular asset, then the most pessimistic investor, 1, must be short-selling constrained for that asset. Further, note that the most pessimistic investor must also have the smallest demand for the other asset, since they are the most pessimistic for *both* assets. Thus, to ascertain whether *some* investor type is short-selling constrained, it is enough to check whether investor 1 is short-selling constrained for either asset without checking for any other type (Figure 3, top left).

Suppose we reject all slack constraints. We can then check for an equilibrium where investor 1 (the most pessimistic) is a constrained short-seller for asset 1, asset 2 or both. If those cases are rejected, we can check if investors 1 and 2 are constrained for asset 1, asset 2 or both, bearing in mind that investor 1 must be constrained for asset i if investor 2 is.

This gives us another three cases (investor 2 is constrained for asset 1, for asset 2, or both), and the first two can happen two ways: investor 1 constrained for both assets, or investor 1 constrained only for the asset investor 2 is constrained for – giving an extra five cases.

We can thus count all cases up to $K-1$ short-selling constrained types, as in Figure 3. The number of cases given $\tilde{k}$ constrained types is $2\tilde{k}+1$, so the total number of cases is $1(0 \text{ constrained})+3$ (1 constrained) $+5$ (2 constrained)$+...+2K-1$ ($K-1$ constrained) $= K^2$. Hence, the 'aligned optimism' assumption reduces the number of cases to check from $4^K-(2^{K+1}-1)$ to the quadratic K^2, which is computationally tractable; see Corollary 2.

Corollary 2 *Let $\tilde{P}_{t,(h)}$ be the price vector when investor types $1,...,h$ are short-selling constrained for both assets and types $h+1,...,K$ are slack for both assets, and let $\tilde{P}_{t,(h',h)}$ be the price vector when some types $h',...,h$ are constrained for one asset only (where $1 \leq h' \leq h$).*

(i) If $\exists \tilde{k} \in \{0,1,...,K-1\}$ such that $\overline{\mathbf{V}}_i(t)(\mathbb{F}_t^{\tilde{k}} - \hat{\mathbb{F}}_t + \hat{N}_t) < 0 \leq \overline{\mathbf{V}}_i(t)(\mathbb{F}_t^{\tilde{k}+1} - \hat{\mathbb{F}}_t + \hat{N}_t)$ for $i=1,2$, then types $1,..,\tilde{k}$ are short-selling constrained for both assets (none if $\tilde{k}=0$), types $\tilde{k}+1,...,K$ buy both assets, and the equilibrium price vector and demands are

$$\tilde{P}_t = \frac{1}{\sum_{k=\tilde{k}+1}^{K} n_t^k}\left(\sum\nolimits_{k=\tilde{k}+1}^{K} n_t^k \mathbb{F}_t^k - \left[\overline{\mathbf{V}}(t)\right]^{-1}\overline{N}\right) := \tilde{P}_{t,(\tilde{k})} \tag{26}$$

$$N_{i,t}^k = 0 \quad \forall k \leq \tilde{k},\, i=1,2, \qquad N_{i,t}^k = \overline{\mathbf{V}}_i(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \quad \forall k > \tilde{k},\, i=1,2 \tag{27}$$

where $\hat{\mathbb{F}}_t = \frac{1}{\sum_{k=\tilde{k}+1}^{K} n_t^k}\sum_{k=\tilde{k}+1}^{K} n_t^k \mathbb{F}_t^k$ and $\hat{N}_t = \frac{1}{\sum_{k=\tilde{k}+1}^{K} n_t^k}[\overline{\mathbf{V}}(t)]^{-1}\overline{N}$.

(ii) Else, some types are constrained for one asset only and $\exists\ \tilde{k} \in [K-1]$, $\hat{k} \in [1,\tilde{k}]$ s.t. for all $i \in [2]: \overline{\mathbf{V}}_i(t)(\mathbb{F}_t^{\hat{k}-1} - \hat{\mathbb{F}}_t + \hat{N}_t) < 0 \leq \hat{\mathbf{V}}_i(t)(\mathbb{F}_t^{\tilde{k}+1} - \hat{\mathbb{F}}_t + \hat{N}_t)$, and for all $h \in \{\hat{k},...,\tilde{k}\}$:

$$\exists i \in [2]: \overline{\mathbf{V}}_i(t)(\mathbb{F}_t^h - \hat{\mathbb{F}}_t + \hat{N}_t) < 0 \leq \overline{\mathbf{V}}_{-i}(t)(\mathbb{F}_t^h - \hat{\mathbb{F}}_t + \hat{N}_t),\ \mathbb{F}_{-i,t}^h \geq \hat{\mathbb{F}}_{-i,t} - \hat{N}_{-i,t} \tag{28}$$

and the corresponding equilibrium price vector and demands for $i=1,2$ are

$$\tilde{P}_t = \tilde{\mathbf{V}}_t^{-1}\left(\sum\nolimits_{k=\hat{k}}^{\tilde{k}} n_t^k \hat{\mathbf{V}}(t)\mathbb{F}_t^k + \sum\nolimits_{k=\tilde{k}+1}^{K} n_t^k \overline{\mathbf{V}}(t)\mathbb{F}_t^k - \overline{N}\right) := \tilde{P}_{t,(\hat{k},\tilde{k})} \tag{29}$$

$$\forall k < \hat{k}: \ \ N_{i,t}^k = 0, \qquad \forall k > \tilde{k}: \ \ N_{i,t}^k = \overline{\mathbf{V}}_i(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \tag{30}$$

$$\forall k \in [\hat{k},\tilde{k}]: \ \ N_{1,t}^k, N_{2,t}^k = \begin{cases} \hat{\mathbf{V}}_1(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0,\ 0 & \text{if (28) holds for } i=1 \\ 0,\ \hat{\mathbf{V}}_2(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 & \text{if (28) holds for } i=2 \end{cases} \tag{31}$$

where $\hat{\mathbb{F}}_t = \tilde{\mathbf{V}}_t^{-1}\left(\sum_{k=\hat{k}}^{\tilde{k}} \hat{\mathbf{V}}(t)\mathbb{F}_t^k + \sum_{k=\tilde{k}+1}^{K} \overline{\mathbf{V}}(t)\mathbb{F}_t^k\right)$, $\hat{\mathbf{V}}_i(t)$ has a single non-zero entry $\hat{v}_{ii}(t) = \frac{|\overline{\mathbf{V}}(t)|}{\overline{v}_{-i-i}(t)} > 0$, $\hat{N}_t = \tilde{\mathbf{V}}_t^{-1}\overline{N}$, and $\tilde{\mathbf{V}}_t = \sum_{k=\hat{k}}^{\tilde{k}} n_t^k \hat{\mathbf{V}}(t) + \sum_{k=\tilde{k}+1}^{K} n_t^k \overline{\mathbf{V}}(t)$.

Proof. It follows from Proposition 1 – see the Supplementary Appendix. ■

Corollary 2 sets out the different types of equilibrium possible when beliefs satisfy (25); recall that there is a unique equilibrium among these cases. For the cases in Part (i), any short-selling constrained types are out of *both* markets. Part (ii) is more complicated: some types $\hat{k}, ..., \tilde{k}$ are short-selling constrained for one asset but not the other, where $\tilde{k}$ is the number of investors short-selling constrained for *some* asset. Condition (28) says that *one* of the assets is held by some types $\hat{k}, ..., \tilde{k}$ (here $1 \leq \hat{k} \leq \tilde{k}$) while the other asset is *not*.

Although (25) assumes inverse covariance matrices are identical, scaled covariance matrices or differences in risk aversion can easily be dealt with numerically. To see why, let $\mathbf{V}^k(t) = c^k \overline{\mathbf{V}}(t)$, where $c^k > 0 \ \forall k$. Then by a similar argument to that in Section 4.1.2, an algorithm based on ordering by optimism will separate types into sets of unconstrained or constrained for each asset, which is the key requirement for finding a fast solution.[11]

4.2.2 Negative alignment

We complete our study of aligned optimism by considering negative alignment. In this case, investors *disagree* in their optimism towards assets 1 and 2. We refer to this case as *negative* alignment because the optimism ranking for one asset as *inverted* for the other asset. This case is useful for studying sharply different investor attitudes across assets (e.g. our Example 1 above) and it also facilitates the study of trend-following investors when both overvalued and undervalued assets can exist simultaneously in a financial market, e.g. a stock exchange.

For identical covariance matrices ($\mathbf{V}^k(t) = \overline{\mathbf{V}}(t), \ \forall k$) the negative alignment condition is

$$\begin{aligned} &\mathbb{F}^1_{1,t} < ... < \mathbb{F}^K_{1,t}, \qquad && \overline{\mathbf{V}}_1(t)\mathbb{F}^1_t < \ldots < \overline{\mathbf{V}}_1(t)\mathbb{F}^K_t \\ &\mathbb{F}^K_{2,t} < ... < \mathbb{F}^1_{2,t}, \qquad && \overline{\mathbf{V}}_2(t)\mathbb{F}^K_t < \ldots < \overline{\mathbf{V}}_2(t)\mathbb{F}^1_t \end{aligned} \tag{32}$$

where investors $k = 1, \ldots, K$ are ordered according to their optimism for asset 1.

Intuitively, if views about two assets are polarized, then the investor most optimistic about one asset may be the least optimistic about the other asset, as in (32). An example would be the comparison of a cryptocurrency asset and a gold-based asset; investors typically have very strong views about these assets, and it does not seem uncommon for those most bullish about the gold-based asset – which has some fundamental value – to be highly sceptical (i.e. pessimistic) about a crypto-based asset that has no intrinsic value.

Negative alignment also applies if investors are trend followers and one asset is overvalued relative to its fundamental price while the other asset is undervalued. In particular, if investors differ in the strength of trend following but have identical extrapolation coefficients for both assets, then the strongest trend follower will simultaneously be the most optimistic for the overvalued asset and the most pessimistic for the undervalued asset, while the weak-

[11] The inequalities are $\max_{k\in\{1,...,\tilde{k}_i\}} \left(c^k \overline{\mathbf{V}}_i(t)(\mathbb{F}^k_t - \tilde{P}_t)\right) < 0 \leq \min_{k\in\{\tilde{k}_i+1,...,K\}} \left(c^k \overline{\mathbf{V}}_i(t)(\mathbb{F}^k_t - \tilde{P}_t)\right)$, $i = 1, 2$, where $\tilde{k}_i \in \{1, ..., K-1\}$ is the number of short-selling constrained investors for asset i; compare to (24). Note that no investors will be short-selling constrained for asset i if $\min_{k\in\{1,...,K\}} \left(c^k \overline{\mathbf{V}}_i(t)(\mathbb{F}^k_t - \tilde{P}_t)\right) \geq 0$. We study scaled covariance matrices numerically in the final section of the paper (see Section 5.4).

est trend follower will be in the opposite situation (with intermediate extrapolators falling between these two extremes). We consider such an example in Section 5.3.

As noted, the beliefs in Example 1 satisfy (32), and observe that our definition of optimism here applies to price beliefs (or forecasts) $\mathbb{F}_t^k$ and the same optimism ranking must be *preserved* for risk-weighted beliefs (i.e. $\overline{\mathbf{V}}(t)\mathbb{F}_t^k$); note that this preservation assumption was also made in the positive alignment case in the previous section. As in the positive alignment case, we can characterize the equilibrium price vector and demands in terms of active-slack conditions on beliefs, and it can be shown that the total number of cases is K^2. These formal results, which are similar to those for the positive alignment case, are provided in the Supplementary Appendix (here) and allow us to design a fast solution algorithm.

4.3 Discussion

We have set out three cases where large-scale heterogeneity can be studied while avoiding the computational intractability of the curse of dimensionality. These cases are economically and empirically appealing given investor behaviour and short-selling regulations observed in practice. A key insight used in our approach is that investors cannot be buyers if more optimistic investors are not. This insight allows us to rule out many cases that would otherwise be checked, thereby streamlining the search for an equilibrium. Our results have wide applicability, and we provide solution algorithms in the Supplementary Appendix.

Our results use methods that have been applied to the case of one short-selling constrained asset (Anufriev and Tuinstra, 2013; Hatcher, 2024b,a), but we take on the difficult case of *two* short-selling constrained assets whose prices are jointly determined. Unlike some previous works that studied static models with short-selling constraints (Tong et al., 2017a,b; Ma et al., 2018), our approach allows the study of price dynamics, and our solution algorithms (for the two asset case) are fast and avoid the use of approximation. Our dynamic pricing approach is in the spirit of Brock et al. (2005), who study asset pricing under evolutionary competition with an arbitrary number of predictors. However, while they study one risky asset and abstract from short-selling constraints, we study two risky assets and add short-selling constraints to their many-predictor world (i.e. large-scale heterogeneity).

The usefulness of these results is illustrated by our numerical application in the next section, where many predictors compete in evolutionary competition, subject to short-selling constraints. To solve the numerical applications in an efficient manner, we built three solution algorithms based on our analytical results; Algorithms 1, 2 and 3 can be found in the Supplementary Appendix, and the codes are available on the author's Github page.[12]

When presenting our analytical results we assumed that investors have identical (inverse) covariance matrices for simplicity; however, as indicated in sections 4.1.2 and 4.2.1–4.2.2, this assumption can be relaxed for the purpose of computation, so that investors may differ in risk aversion or the scaling of covariances. Moreover, in the case of one short-selling

[12]See the repository at https://github.com/MCHatcher/Jarrow-model-dynamic.

constraint, it is enough that the inverse covariance matrices of investors share a single row in common (the one relating to the short-selling constrained asset); as a result, investor types can differ in their underlying covariance matrices $\Omega^k(t)$ in this case. By considering both positive and negative alignment, we allow investors to be consistent in their optimism across assets or to have sharply differing views about each asset's value.

For analytical convenience we have assumed that beliefs can be unambiguously ordered, such that optimism ties are excluded and our analytical results can be neatly stated. Importantly, ties in terms of optimism do not pose any problems from a computational perspective, and such ties can at any rate be eliminated via minor changes in the underlying beliefs. Also, note that while investors with distinct price forecast vectors $\mathbb{F}_t^k$ may have identical risk-weighted price beliefs $\overline{\mathbf{V}}(t)\mathbb{F}_t^k$, such investors are heterogeneous only in a weak sense.

5 Numerical application

5.1 Model

We consider a two-asset extension of the Brock and Hommes (1998) model with many heterogeneous types $k \in [K]$. Risky assets $i \in \{1, 2\}$ are in positive net supply and pay a stochastic dividend $D_{i,t} = \overline{D}_i + \varepsilon_{i,t}$ at date t, where $\varepsilon_{i,t}$ is a mean-zero innovation with variance $\sigma_{d_i}^2$ and $corr(\varepsilon_{1,t}, \varepsilon_{2,t}) \in (-1, 1)$. All types have identical beliefs about covariances and equal risk aversion, $a^k = 1$. The riskless asset (in flexible supply) pays a fixed return $r > 0$ per period.

The beliefs of types $k \in [K]$ are given by

$$\mathbb{E}_t^k P_{t+1} = P^f + g^k(P_{t-1} - P^f) \tag{33}$$

$$\mathbf{\Omega}^k(t) = \overline{\mathbf{\Omega}}, \qquad \forall t, k \tag{34}$$

where P^f is the fundamental price vector, $g^k \geq 0$ is a trend-following parameter, and $\overline{\mathbf{\Omega}}$ is a common covariance matrix whose off-diagonal entries reflect the sign of $corr(\varepsilon_{1,t}, \varepsilon_{2,t})$.

We model two broad groups of investors: chartists and fundamentalists. If $g^k = 0$, then k is pure fundamentalist, while if $g^k > 0$, k is a chartist (or trend-follower). Within these broad groups, investors are heterogeneous. The fundamental price is

$$P^f = \frac{1}{r}\left(\overline{D} - \overline{\mathbf{\Omega}}\,\overline{N}\right) \tag{35}$$

where $\overline{N}$ is the vector of positive outside supplies, $r > 0$ is the riskless rate, and we choose $\overline{D}$ so the fundamental prices are positive; note that when deriving (35), we assume common rational expectations and impose a no-bubbles condition when solving forwards.[13]

[13]To derive the fundamental price, either set $K = 1$, $n_t^1 = 1$ (representative rational investor) or $K > 1$, $n_t^k = 1/K\ \forall k$ (K rational types). Demands are $N_t^k = (\overline{\mathbf{\Omega}})^{-1}(\mathbb{E}_t^k P_{t+1}^f + \mathbb{E}_t^k D_{t+1} - (1+r)P_t^f)$. With common conditional expectations, $\mathbb{E}_t^k P_{t+1}^f = \mathbb{E}_t[P_{t+1}^f]$, $\mathbb{E}_t^k D_{t+1} = \mathbb{E}_t[D_{t+1}] = \overline{D}$ and market clearing yields (35).

The demands of type k are given by

$$\begin{aligned} N_t^k &= \hat{\mathbf{V}}^k(t)(\mathbb{E}_t^k P_{t+1} + \mathbb{E}_t^k D_{t+1} - (1+r)P_t) \\ &= \hat{\mathbf{V}}^k(t)(P^f + \overline{D} + g^k(P_{t-1} - P^f) - (1+r)P_t) \\ &= \hat{\mathbf{V}}^k(t)(\mathbb{F}_t^k - (1+r)\tilde{P}_t) \end{aligned} \tag{36}$$

where $\mathbb{F}_t^k = \overline{\mathbf{\Omega}}\,\overline{N} + g^k \tilde{P}_{t-1}$, $\tilde{P}_t = P_t - P^f$ is the *deviation* from the fundamental price, and

$$\hat{\mathbf{V}}^k(t) = \begin{cases} \overline{\mathbf{V}} := \overline{\mathbf{\Omega}}^{-1} & \text{if } \overline{\mathbf{V}}_1(\mathbb{F}_t^k - (1+r)\tilde{P}_t), \overline{\mathbf{V}}_2(\mathbb{F}_t^k - (1+r)\tilde{P}_t) \geq 0 \\ \begin{bmatrix} \frac{|\overline{\mathbf{V}}|}{\overline{v}_{22}} & 0 \\ 0 & 0 \end{bmatrix} & \text{if } \mathbb{F}_{1,t}^k \geq (1+r)\tilde{P}_{1,t},\ \overline{\mathbf{V}}_2(\mathbb{F}_t^k - (1+r)\tilde{P}_t) < 0 \\ \begin{bmatrix} 0 & 0 \\ 0 & \frac{|\overline{\mathbf{V}}|}{\overline{v}_{11}} \end{bmatrix} & \text{if } \mathbb{F}_{2,t}^k \geq (1+r)\tilde{P}_{2,t},\ \overline{\mathbf{V}}_1(\mathbb{F}_t^k - (1+r)\tilde{P}_t) < 0 \\ \mathbf{0} & \text{if } \mathbb{F}_{i,t}^k < (1+r)\tilde{P}_{i,t} \text{ for } i = 1,2 \end{cases} \tag{37}$$

where the middle cases have investor type k short-selling constrained for one asset only; note that $\hat{\mathbf{V}}^k(t) = \overline{\mathbf{V}}$ holds for all t and k if short-selling is unrestricted.

The population shares of different types are determined by *evolutionary competition* between alternative predictors (Brock and Hommes, 1997, 1998). In particular, we take the past return on the investment portfolio (net of a predictor cost) as our performance measure:

$$U_t^k = \sum_{i=1}^{2} R_{i,t} N_{i,t-1}^k - C_k \tag{38}$$

where U_t^k is realized trading profit of type k given a predictor cost $C_k \geq 0$, and for $i = 1, 2$

$$\begin{aligned} R_{i,t} &= P_{i,t} + D_{i,t} - (1+r)P_{i,t-1} \\ &= \tilde{P}_{i,t} - (1+r)\tilde{P}_{i,t-1} + (\overline{\mathbf{\Omega}}\,\overline{N})_i + \varepsilon_{i,t}. \end{aligned} \tag{39}$$

Population shares are updated via a logistic model with intensity of choice $\beta \in [0, \infty)$:

$$n_{t+1}^k = \frac{\exp(\beta U_t^k)}{\sum_{k \in [K]} \exp(\beta U_t^k)}, \qquad \forall k \in [K]. \tag{40}$$

The intensity of choice determines the willingness of investors to switch to better-performing predictors. In the special case $\beta = 0$ no switching occurs; increasing β implies more switching to relatively profitable predictors. Note the population share of each type depends on the *joint* investment performance of the predictor across *both* assets, so population shares are not asset-specific, though this assumption can be relaxed (see the Supplementary Appendix).

We set some parameters at common values in all simulations. We normalize asset outside

supplies to 1 and expected dividends to 10, i.e. $\overline{N} = \begin{bmatrix}1 & 1\end{bmatrix}'$ and $\overline{D} = \begin{bmatrix}10 & 10\end{bmatrix}'$. The dividend shocks $\varepsilon_{i,t}$ are drawn from a multivariate-Normal distribution $\mathcal{N}(\mathbf{0}, \frac{1}{100^2}\overline{\mathbf{\Omega}})$; we give $\overline{\mathbf{\Omega}}$ diagonal entries 1 and 3 and off-diagonal entry $\rho_{12}\sqrt{3}$, so that $\rho_{12} \in (-1, 1)$ controls the subjective correlation in returns (we assume $sgn(\rho_{12}) = sgn(corr(\varepsilon_{1,t}, \varepsilon_{2,t}))$.[14] We set the riskless interest rate at $r = 0.01$ (or 1%). The other parameters differ across our examples.

As mentioned, types within the fundamentalist and chartist groups have heterogeneous price beliefs. We assume each group is of equal size $K/2$, which varies in our examples.

5.2 Short-selling ban on one asset

We start with the relatively simple case where there is a short-selling ban on one asset only. We take asset 2 as subject to a short-selling ban, as in Section 4.1; recall that the number of cases is linear in K for this problem, which can be solved using our Algorithm 1.

The fundamentalist group of investors have trend-following parameters g^k close to zero, while chartists have much larger trend parameters (more than four times as large as any fundamentalist). The chartist predictor is costless ($C^k = 0$) whereas the fundamental predictor has cost $C^k = 1 - g^k$; in particular, the fundamentalists have trend parameters linearly spaced on $[0, 0.20]$ and chartists have trend parameters linearly spaced on $[0.80, 1.10]$. We set the intensity of choice at $\beta = 0.975$, the initial price vector at $\tilde{P}_0 = \begin{bmatrix}2 & 8\end{bmatrix}'$, and the correlation in returns at $\rho_{12} = -0.65$ (i.e. the assets are viewed as hedges by investors).

Table 1: Computation time and accuracy for various K ($T = 50$ periods)

No. of types, K	Time (s)	Bind freq.	$\max(Error_t)$
100	0.01	23/50	1.3e-15
1,000	0.01	19/50	3.1e-15
10,000	0.07	18/50	8.1e-14
100,000	0.52	18/50	5.0e-13
1 million types	6.94	18/50	5.2e-14
1 million - Unrestricted short selling	4.64	-	1.2e-15

Notes: $\max(Error_t) = \max_{t\in[T]}\left(|\sum_{k=1}^{K} n_t^k N_t^k - \overline{N}|\right)$. N_t^k = demands, n_t^k = pop. shares.

We first look at the performance of Algorithm 1 in solving the model (Table 1). The simulations times are fast and the maximum discrepancy in the market-clearing condition is essentially zero over all 50 periods.[15] Short-selling constraints bind for some type in around the first 20 periods (Column 3). As expected, solution times increase with the number of investor types K, being close to linear for very large K values. For $K = 10,000$ types the

[14] As is standard in the literature, we normalize the subjective variances by setting $\sigma_{11} = 1$ and choose relatively small variances for dividend shocks – see e.g. Anufriev and Tuinstra (2013) in a one-asset setting. Our choice for $\overline{\mathbf{\Omega}}$ is also similar to Examples 1–2, but with a negatively signed covariance (see Section 5.2).

[15] Simulations were run in MATLAB 2025a (Windows version) on a Dell OptiPlex Desktop with an Intel Core i5-13500 2.50 GHz processor and 16GB of RAM. No parallel processing (parfor) is used in our codes.

solution takes around 0.1 seconds; for $K = 100,000$ about half a second. Solving the model with 1 million types takes just under 7 seconds. The simulation errors in Column 4 are very small but not exactly zero; the reason seems to be that market-clearing sums many demands scaled by population shares. There is also a very small simulation error under unrestricted short-selling (final row), for which the well-known analytical solution is used in our code.

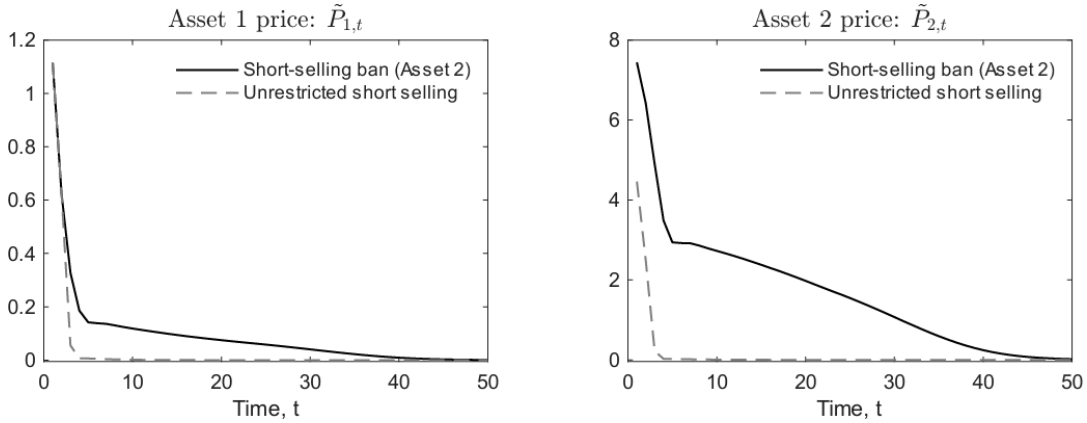


Figure 4: Equilibrium asset prices for $K = 100$ types (ban vs no ban)

Figure 4 plots the asset price dynamics with and without a short-selling ban on Asset 2 when there are $K = 100$ types. The short-selling ban raises the price of asset 2 substantially. The price of asset 2 initially falls sharply regardless of whether short-selling is banned or permitted; however, while the price (in deviations from the fundamental price) quickly falls towards zero under unrestricted short-selling (dashed line), the price fall is briefly reversed under a short-selling ban, and we then see a period in which the price deviation is persistent and converges slowly on zero. The price dynamics differ both qualitatively and numerically.

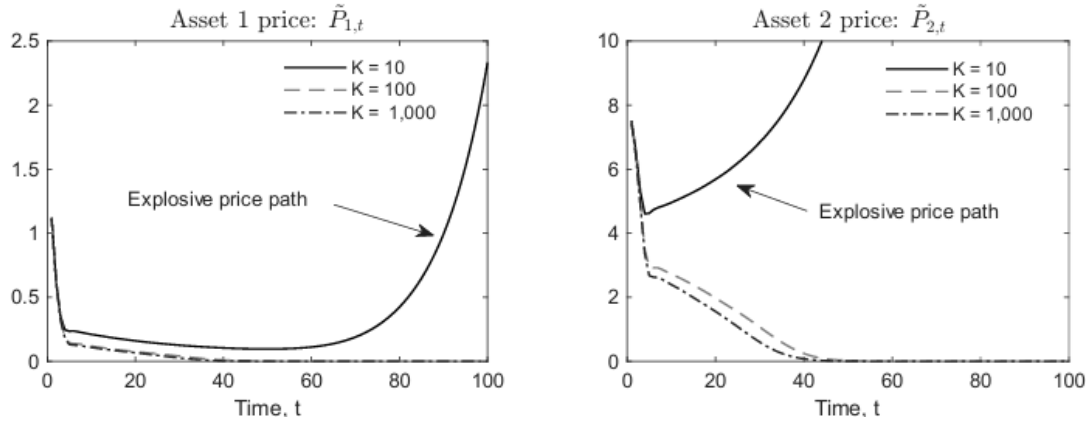


Figure 5: Equilibrium price paths as the number of types K is increased (ban)

Intuitively, a short-selling ban on asset 2 not only raises its price, but affects trading profitability since the constrained (fundamental) types have a zero position in asset 2 and so perform worse than if they could short sell and benefit from price falling; as a result, the population shares of trend followers are higher, making (weighted) price beliefs more persistent and thus asset prices also. At the same time, a similar difference in persistence is seen for asset 1, which makes sense given that the asset prices are jointly determined and a larger weight on the trend also corresponds to more persistent beliefs in the asset 1 price.

Figure 5 gives a comparison of the price paths as K is increased. These results illustrate that a small number of heterogeneous investors need not provide a good approximation to the equilibrium in a many-investors world: for $K = 10$ the price paths are explosive, while for the other cases in Table 1 ($K = 100$ types and higher), increasing K makes the prices (in deviations from the fundamental prices) converge to zero. In all plots, the sequence of dividend shocks (which affects realized profits and thus type updating) is identical.

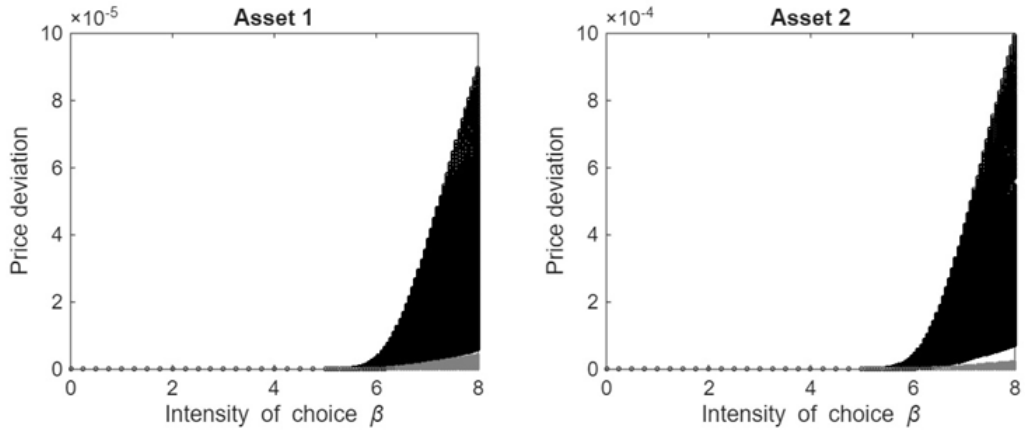


Figure 6: Numeric bifurcation diagrams for $K = 100$ types: short-selling ban (black) and unrestricted short selling (grey). For each value of β, we plot 60 points following a deterministic transitory of 990 periods from initial price deviations $\tilde{P}_{1,0} \in (1, 3)$, $\tilde{P}_{2,0} \in (7, 9)$.

Finally, we give a bird's eye view of the price dynamics by simulating a bifurcation diagram as the intensity of choice parameter β is increased (see Figure 6); here we consider the *deterministic skeleton* by setting dividend shocks at zero and use a slightly narrower range of trend-following parameters ($[0.80, 1.073]$) to prevent frequent explosive simulations. For small enough values of β, price trajectories generally converge to the fundamental steady state price vector, P^f. However, once the intensity of choice is large enough, there is a bifurcation and (quasi-)periodic price orbits; it is clear from Figure 6 that these price fluctuations are larger with a short-selling ban (black) than under unrestricted short selling (grey), and both assets are overvalued. Further, for a short-selling ban we observed some explosive price paths. For each case, the bifurcation results took just over one minute to compute.[16]

[16]For unrestricted short selling we did not see any explosive price paths; for a short-selling ban around 1

Overall, the impact of a short-selling ban is mixed. A short-selling ban can lead to explosive price paths and we saw more persistence in asset mispricing when short-selling is prohibited. On the other hand, a sharp drop in asset prices is prevented by a short-selling ban, along with the associated poor performance of trend followers (who expect high initial prices to remain). A regulation that bans short selling therefore has implications for distribution across investor types who follow different trading strategies.

5.3 Short-selling ban on both assets

We now consider a short-selling ban on both assets, as in Section 4.2. Thus, we rely on Algorithms 2 and 3 here, which we expect to be much slower than Algorithm 1, as number of cases is *quadratic* in the number of types. The model itself is unchanged relative to the previous section, except that short-selling is prohibited for both risky assets; hence the demand schedules follow (36) and the optimism of types depends on the predictors (33)–(34).

In the examples below, expectations are positively aligned at date t if both inherited price deviations $\tilde{P}_{i,t-1}$ have the same sign, while for opposite signs, beliefs are negative aligned and the optimism ranking is flipped for each asset; see (33); hence, by varying the signs we can consider different cases (positive vs negative alignment) and apply our algorithms. Thus, our model with fundamentalists and chartists has *aligned optimism* as in Sections 4.2.1–4.2.2.[17]

We start out with initial price deviations of $\tilde{P}_0 = \begin{bmatrix} 2 & 8 \end{bmatrix}'$ as above, which implies that beliefs are positively aligned. A small change is that we choose the trend parameters as linearly spaced on the narrower interval $[0.80, 1.09]$ to prevent generally explosive paths.

Table 2: Computation time and accuracy for various K ($T = 50$ periods)

No. of types, K	Time (s)	Bind freq.	$\max(Error_t)$
10	0.03	50/50	5.8e-15
100	0.20	50/50	3.6e-15
$1,000$	3.38	50/50	4.4e-15
$2,000$	9.97	50/50	4.0e-15
$4,000$	32.96	50/50	6.1e-15
$4,000$ - Unrestricted short selling	0.03	-	5.6e-16

Notes: $\max(Error_t) = \max_{t\in[T]} \left(|\sum_{k=1}^{K} n_t^k N_t^k - \overline{N}| \right)$. N_t^k = demands, n_t^k = pop. shares.

Table 2 looks at the performance of Algorithm 2 in solving the model. Computation times are much slower than in Table 1 as expected, and the scaling as the number of investors K is increased looks close to quadratic for very high numbers of types. The maximum simulation errors across all 50 periods are again very small but not exactly zero, as in Table 1. Note that

in 4 simulations was explosive at higher values of β (we terminate since once an 'inf' of 'NaN' was observed).

[17] In general, the price deviation may change sign during a simulation if it is subject to shocks or if types differ in their 'bias' (or intercepts) which may be positive or negative. In the simple example here the bias is equal to zero for all types, such that the heterogeneity in beliefs comes from the trend parameters g^k alone.

Algorithm 3 could also be used to solve for the simulation paths in this positive alignment example; it delivers identical numerical solutions and very similar computation times.

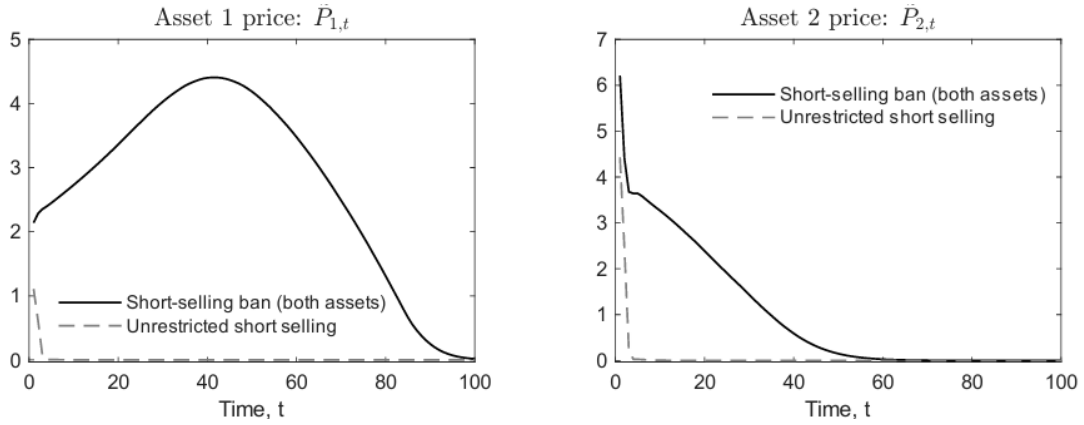


Figure 7: Equilibrium price paths: $K = 100$, short-selling ban on both assets

In Figure 7 we plot the simulated price paths for $K = 100$ types and $T = 100$ periods. Asset 1 is overvalued (a positive deviation from the fundamental price) and its price starts on an upward trajectory before declining, hence there is a price bubble that collapses, with the overvaluation converging to zero. By comparison, the price of asset 2 declines but is far more persistent than if short-selling were unrestricted, similar to Figure 4 where only short selling of Asset 2 was prohibited. If short-selling were unrestricted for both assets, the prices of both assets decline monotonically (see dashed lines) and the price of asset 1 converges much faster; hence, the bubble trajectory is induced by the profit gain to trend followers, which is quite intuitive. Note that a shorting ban on asset 2 only (not shown) delivers similar results to Figure 4, thereby reducing mispricing somewhat relative to a ban on both assets.

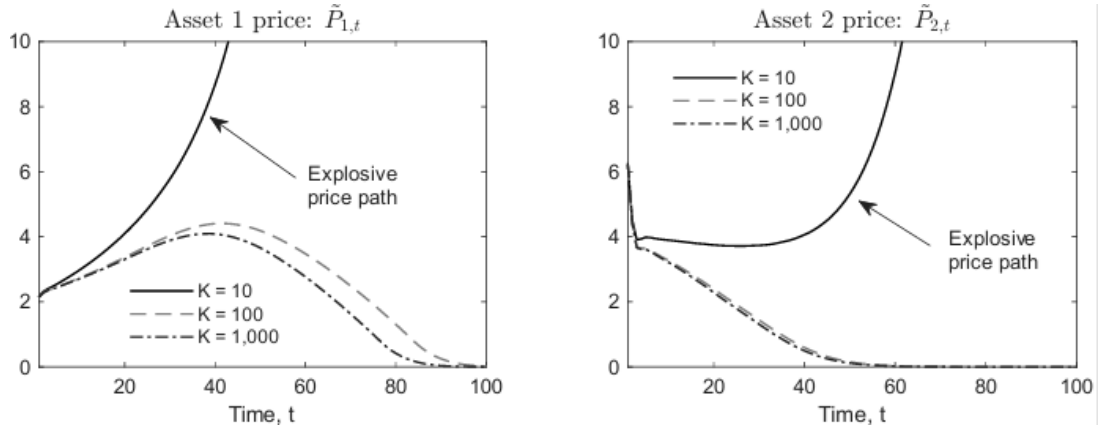


Figure 8: Equilibrium price paths as the number of types K is increased (ban)

Figure 8 plots the price dynamics for different investor types, K, when short-selling is prohibited. As in Figure 5 we see that using relatively small numbers of types does not provide a good approximation to the price dynamics; for example, the price paths are explosive with just 10 types. Further, increasing the number of types not only implies price stability but also reduces the amplitude and persistence of the price bubble (dashed lines).

As a final example, we consider the case of *negative alignment*, i.e. investors who have opposite opinions about the two risky assets. We therefore assign initial price deviations that differ in sign, by setting $\tilde{P}_0 = \begin{bmatrix} 7 & -12 \end{bmatrix}'$, such that trend followers are optimistic for asset 1 (overpriced) and pessimistic for asset 2 (undervalued). In addition, we reduce the negative correlation of the asset returns to $-1/3$ to ensure the negative alignment condition is satisfied, and we reduced the intensity of choice β, as this helps to visually illustrate the results (see below).[18] The computation times and accuracy for Algorithm 3 (negative alignment) are similar to those in Table 2 and are provided in the Supplementary Appendix.

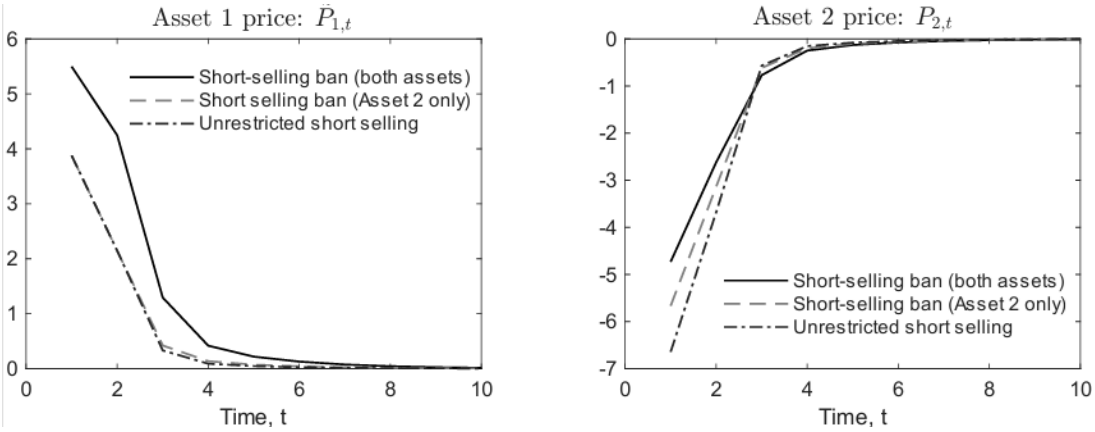


Figure 9: Equilibrium price paths for $K = 100$: short-selling bans (2 assets; 1 asset; no ban)

In Figure 9 we plot the price paths for $K = 100$ types and three cases: a short-selling ban on one asset (asset 2), a short-selling ban on *both* risky assets, and unrestricted short selling. The short-selling ban on both assets raises both prices, exacerbating the overvaluation of asset 1 while diminishing the undervaluation of asset 2. A shorting ban on asset 2 only raises its price around half as much as a two-asset ban, but the asset 1 price is barely affected, giving much lower overvaluation and faster price convergence, as for unrestricted short-selling. Comparing across cases, the one-asset ban performs well on overall mispricing because it reduces the overvaluation of asset 1 while substantially diminishing the undervaluation of asset 2. These results point to a role for selective shorting-selling bans as opposed to a blanket policy of banning shorting of all shares for some time; more generally, a theme that emerges from our results is the need for careful policy analysis in a multi-asset setting.

[18] We reduced the intensity of choice from 0.975 to 0.125; the trend parameters are spaced on $[0.805, 1.090]$.

5.4 Heterogeneous covariances

In all the examples thus far, we considered homogeneous covariances and risk aversion, as seen in (34) and (37). However, as discussed in sections 4.1.2 and 4.2.1–4.2.2, the assumptions of common covariances and equal risk aversion can be relaxed. For the examples above, we also allowed different risk aversion coefficients for each investor (or equivalently, differently scaled covariance matrices) by setting $\mathbf{V}^k(t) = c^k\overline{\mathbf{V}}(t) = c^k\overline{\mathbf{V}}$ with $c^k = 1$ (normalization).

The computation times and accuracy are similar to Tables 1–2, but with somewhat longer simulation times as expected. We provide these additional numerical results in the Supplementary Appendix. As well as scaling of covariances (or risk aversion), we present results for the case where inverse covariance matrices $\mathbf{V}^k$ differ in general but have the same second row; this extension relates to one short-selling constrained asset (see Section 4.1.2).

6 Conclusion

We have studied equilibrium asset pricing when investors are heterogeneous and face regulations that prohibit short selling. This is a tricky fixed-point problem because the active-slack decision of each investor depends on the market price, and the price in turn depends on how many investors are short-selling constrained. It is well-known that such problems are, in general, both analytically and computationally intractable due to the large number of distinct cases in which constraints may be active; these problems are amplified in a dynamic model as here where an equilibrium must be found in each period. In this paper we proposed a way forward for the case of two risky assets and an arbitrary number of investor types whose population shares can depend on past performance via evolutionary competiton.

We provided conditions on beliefs such that short-selling constraints are active or slack in equilibrium for any number of investors and – using these results – we presented two scenarios of economic interest: (i) a short-selling ban on one asset, (ii) aligned optimism among heterogeneous investor types. In both these scenarios, our active-slack analytics make it computationally tractable to find a solution. As a result, the consequences of short-selling constraints can be studied absent the usual curses of dimensionality that come from solving a high-dimensional problem that involves many investors and many short-selling constraints.

We made use of the insight that ordering K investors in terms of their optimism allows many cases to be ruled out *a priori*. As a result, we get down to K cases for a short-selling ban on one asset, and K^2 cases for aligned optimism, as compared to exponential in K when beliefs are unsorted. We designed fast and accurate solution algorithms using analytical results, as showcased in our numerical applications. Our numerical examples also shed light on potential consequences of regulations that ban short-selling, and an interesting finding was that a selective ban may reduce mispricing compared to a blanket ban on both assets.

In terms of future research, there is much to do as our results apply to two risky assets and impose some restrictions on investor beliefs. An open question is whether larger models with several assets can be solved quickly without making strong simplifying assumptions.

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Appendix

Proof of Proposition 1

Part (i)

Let us guess that short-selling constraints are slack in both markets (for all K investors). Then $N_{i,t}^k = \mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t)$ for all $k \in [K]$ and $i = 1, 2$. Given this guess, market-clearing yields $\sum_{k=1}^K n_t^k \mathbf{V}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) = \overline{N}$, i.e. $\tilde{P}_t = \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t)\right]^{-1} \left(\sum_{k=1}^K n_t^k \mathbf{V}^k(t)\mathbb{F}_t^k - \overline{N}\right)$, where $\mathbf{V}^k(t)$ is positive definite for all k (so $\sum_{k=1}^K n_t^k \mathbf{V}^k(t)$ is positive definite and invertible). Given $\tilde{P}_t$, the above guess is verified if and only if $\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \; \forall k \in [K]$, $i = 1, 2$, which is equivalent to $\min_{k\in[K]} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t)\right) \geq 0$ for $i = 1, 2$, where $\tilde{\mathbb{F}}_t := \sum_{k=1}^K \overline{\mathbf{V}}^k(t)\mathbb{F}_t^k$, $\overline{\mathbf{V}}^k(t) := \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t)\right]^{-1} n_t^k \mathbf{V}^k(t)$, and $\tilde{N}_t := \left[\sum_{k=1}^K n_t^k \mathbf{V}^k(t)\right]^{-1} \overline{N}$, as stated in Prop. 1.

Part (ii)

If $\exists i \in \{1, 2\}$ such that $\min_{k\in[K]} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{\mathbb{F}}_t + \tilde{N}_t)\right) < 0$, the guess in Part (i) is not verified and some constraints bind. Thus, either $S_{1,t}$ or $S_{2,t}$ may be empty, but not both. If k is a buyer of some asset, $k \in B_{12,t}$ or $k \in B_{11,t}$ or $k \in B_{22,t}$, and $N_t^k = \hat{\mathbf{V}}^k(t)(\mathbb{F}_t^k - \tilde{P}_t)$, where $\hat{\mathbf{V}}^k(t)$ equals $\mathbf{V}^k(t)$ if $k \in B_{12,t}$ and has one non-zero entry, $v_{ii}^k(t) = \frac{|\mathbf{V}^k(t)|}{v_{-i-i}^k(t)}$ if $k \in B_{ii,t}$, where $-i$ is the only member of $\{1, 2\} \setminus \{i\}$. Else, $k \in S_{1,t} \cap S_{2,t}$, and for such k, $N_t^k = \mathbf{0}$.

Note that $k \in B_{ii,t}$ implies $k \in S_{-i,t}$, $k \in B_{12,t}$ implies $k \notin S_{1,t} \cup S_{2,t}$, and $k \notin B_{1,t} \cup B_{2,t}$ implies $k \in S_{1,t} \cap S_{2,t}$. Given guessed sets $B_{ij,t}$ for $i, j \in \{1, 2\}$ and $j \geq i$, the sets of buyers are $B_{1,t}, B_{2,t}$ and the sets of constrained for each asset are $S_{1,t}, S_{2,t}$. Given this guess, market-clearing yields $\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) = \overline{N}$, where $\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)$ is positive definite, so $\tilde{P}_t = \left[\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)\right]^{-1} \left(\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)\mathbb{F}_t^k - \overline{N}\right)$ [note that the matrix $\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)$ is shown to be positive definite in the Online Appendix.]

Given $\tilde{P}_t$ and demands, (12), the above guess is verified iff the following conditions hold for $i = 1, 2$: $\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \tilde{P}_t) \geq 0 \; \forall k \in B_{12,t}$, $\mathbb{F}_{i,t}^k < \tilde{P}_{i,t} \; \forall k \in S_{12,t}$; $\mathbb{F}_{i,t}^k \geq \tilde{P}_{i,t}$, $\mathbf{V}_{-i}^k(t)(\mathbb{F}_t^k - \tilde{P}_t) < 0 \; \forall k \in B_{ii,t}$. [For an empty set the 'if' condition is vacuously true.] First two conditions are equivalent to $\min_{k\in B_{12.t}} \left(\mathbf{V}_i^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t)\right) \geq 0$, $\max_{k\in S_{12,t}} \left(\mathbb{F}_{i,t}^k\right) < \hat{\mathbb{F}}_{i,t} - \hat{N}_{i,t}$ for $i = 1, 2$; latter conditions are equivalent to $\min_{k\in B_{ii,t}} \left(\mathbb{F}_{i,t}^k\right) \geq \hat{\mathbb{F}}_{i,t} - \hat{N}_{i,t}$, $\max_{k\in B_{ii,t}} \left(\mathbf{V}_{-i}^k(t)(\mathbb{F}_t^k - \hat{\mathbb{F}}_t + \hat{N}_t)\right) < 0$ for $i = 1, 2$, where $\hat{\mathbb{F}}_t = \sum_{k\in B_{1,t}\cup B_{2,t}} \overline{\mathbf{V}}^k(t)\mathbb{F}_t^k$ and $\overline{\mathbf{V}}^k(t) = \left[\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)\right]^{-1} n_t^k \hat{\mathbf{V}}^k(t)$, $\hat{N}_t = \left[\sum_{k\in B_{1,t}\cup B_{2,t}} n_t^k \hat{\mathbf{V}}^k(t)\right]^{-1} \overline{N}$, as stated. ■